Lynx and Hare Dynamics
Model, Stability, Numerical Simulation and Inverse Inference

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Abstract

We combine a Rosenzweig–MacArthur model, equilibrium and Hopf analysis, numerical experiments, and historical parameter fitting using least squares and an inverse PINN. The Stan pelt-index compilation is split into 1900–1915 training and 1916–1920 evaluation. Primary held-out RMSE is 3.07/3.04 thousand pelts for least squares and 2.85/4.32 for the PINN-derived ODE (hare/lynx). The comparison is mixed; observation-model and bound sensitivity prevent precise biological recovery or a robust ecological limit-cycle claim. Frozen numerical evidence, sources and all sensitivity variants are retained. No final integration step retunes a model using test-year errors.

Contributions and evaluation chronology. Arnav: model and units. Joel: theory. Rishi: numerical validation. Yash: PINN and integration. Advay: data and independent least squares/evaluation. Historical forecasts use the same dimensional RHS and solver configuration. Advay's independent reproduction followed Yash's published scored iteration, so it is not a new blinded test. Training demonstrations in the combined notebook do not overwrite the frozen forecasts.

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1 Problem and observation convention

The project asks whether a mechanistic predator–prey model can explain recurring hare–lynx fluctuations and support useful predictions beyond the observations used for calibration. This section establishes the equations, their biological assumptions, units and dimensionless form. It supplies the common mathematical foundation for the team's stability analysis, numerical simulations and parameter estimation.

The completed historical iteration uses annual observations labelled 1900–1920. The Stan case study supplies a public transcription of Hudson's Bay Company pelt records in thousands of pelts, with a documented source chain [2]. These measurements are trade records, not population censuses. We interpret $H(t)$ and $P(t)$ as nonnegative hare and lynx abundance *indices*, respectively, in explicitly declared species-specific units. The comparison between a continuous state and an annual pelt total is an observation approximation: trapping effort, capture probability, seasonal aggregation and unmodelled ecology can all affect the relationship.

Advay's frozen data split and the reference configuration use one index unit per thousand pelts for each species [1, 21]. The Stan compilation remains the source for this iteration. The alternative tsibbledata compilation disagrees with it, a documented source limitation; neither series establishes absolute animal abundance, and they are not combined. Set $t = \text{Year} - 1900$ in years. The implemented evaluation design uses 1900–1915 for calibration and 1916–1920 for final scoring. Normalization and parameter selection use training observations only. Forecasts have already been frozen and scored; this section documents their observation convention without changing it. It does not provide additional historical fits.

1.1 Explicit observation model

For year Year_i and $t_i = \text{Year}_i - 1900$, write

$$Y_{H,i} = H(t_i) + \varepsilon_{H,i}, \qquad Y_{P,i} = P(t_i) + \varepsilon_{P,i}. \tag{1}$$

Here Y is the recorded annual pelt count in thousands; H, P are the model's noiseless pelt indices; and ε is observation discrepancy in the same species-specific units. Under the working proportional-index assumption O1,

$$H(t) = c_H N_H(t), \qquad P(t) = c_P N_P(t), \qquad c_H, c_P > 0, \tag{2}$$

where N_H, N_P denote latent animal abundance and the unknown observation factors are constant. We fit the effective equations directly in pelt units: *no observation-scale factor is fitted.* This does not set $c_H = c_P = 1$ or identify actual animal counts. Constant factors are absorbed into the effective coefficients through the transformation derived in Eq. (26).

Each annual total is approximated by a point observation of the state, as in the implemented convention [1, 21]. Thus H, P have units of thousands of pelts, and $\dot{H}, \dot{P}$ have units of thousands of pelts per year. No seasonal observation integral is fitted. A latent-population model with estimated observation factors, or an integral over the trapping season, would represent additional mechanisms but require further assumptions and unknowns. O1 and the point-sample approximation keep the present comparison tractable, at the cost of limited biological interpretation.

The main observation loss compares additive residuals after division by the training maxima $s_H = 77.4$ and $s_P = 59.4$:

$$R_{H,i} = \frac{H(t_i) - Y_{H,i}}{s_H}, \qquad R_{P,i} = \frac{P(t_i) - Y_{P,i}}{s_P}. \tag{3}$$

For the PINN data term the state in this expression is the network approximation; for least squares it is the integrated ODE state. Both methods use the same species scales. These weights

are a declared comparison convention, not measured noise standard deviations. Additive discrepancy notation alone establishes neither Gaussian noise nor independence of observations. The reported log-error sensitivity instead emphasizes relative discrepancies. A separate sensitivity fits the 1900 initial states within $[0.5, 1.5]$ times their observed values [21]; the two checks change different assumptions and remain separate outcomes.

2 From Lotka–Volterra to the chosen model

2.1 The reference model and its limitation

For a first comparison, let

$$\dot{H} = rH - cHP, \qquad \dot{P} = ecHP - mP, \tag{4}$$

where a dot means differentiation with respect to years. The prey grows at rate rH without predators, encounters remove prey at rate cHP, conversion of consumed prey contributes $ecHP$ to predator growth, and predator mortality removes mP. Here $c > 0$ is the encounter coefficient; it is not the saturation parameter a introduced below. The other parameters are positive.

The positive equilibrium is $H_* = m/(ec)$, $P_* = r/c$. To see why the cycles are neutral, rather than simply asserting it, define for positive H, P

$$V = ec\,[H - H_* - H_* \log(H/H_*)] + c\,[P - P_* - P_* \log(P/P_*)]\,. \tag{5}$$

The logarithms have dimensionless arguments. Differentiation gives

$$\dot{V} = (ec - m/H)\dot{H} + (c - r/P)\dot{P} \tag{6}$$

$$= (ecH - m)(r - cP) + (cP - r)(ecH - m) = 0. \tag{7}$$

Each bracket in V has a strict minimum at its equilibrium value and diverges at zero or infinity. Thus nonzero level sets are closed curves in the positive quadrant, surrounding the equilibrium. The vector field has no other positive equilibrium on a level set, so trajectories follow periodic orbits. A perturbation generally moves the state to a different orbit; it does not relax back to one selected cycle. The equilibrium is a centre, not an attractor. Also, predator-free prey grows exponentially without a resource limit. These are reasons to refine the model; they do not mean that Lotka–Volterra cannot oscillate.

2.2 Resource limitation and saturating predation

Let the prey's per-capita net growth decrease linearly from r at low abundance to zero at $H = K$. Multiplication by H gives logistic growth,

$$G(H) = rH(1 - H/K). \tag{8}$$

The carrying capacity K describes the prey-only resource limit under fixed environmental conditions; it is not an absolute ceiling on every trajectory.

Let $g(H)$ be consumption per predator. A simple search-and-handling argument assumes an encounter-limited rate qH while searching and a maximum processing rate a. If the fractions of time searching and handling are $g/(qH)$ and g/a, their sum is one. For $H > 0$,

$$\frac{g}{qH} + \frac{g}{a} = 1 \quad \Longrightarrow \quad g(H) = \frac{aqH}{a + qH} = \frac{aH}{b + H}, \qquad b = \frac{a}{q}. \tag{9}$$

Extend continuously by $g(0) = 0$. This is the Holling type II response [7]. It assumes an effectively constant prey level over the short feeding interval and no predator interference. In

index units it is an effective response law, not a direct measurement of handling time. The parameters have transparent interpretations:

$$g(b) = a/2, \qquad \lim_{H\to\infty} g(H) = a, \qquad g(H) \simeq (a/b)H \quad \text{when } H/b \ll 1. \tag{10}$$

The auxiliary $q = a/b$ has units inverse lynx index per year. It is not an additional parameter to estimate. The linear response would be simpler but cannot represent saturation; a type III response would introduce a different low-prey mechanism that the present project does not establish.

2.3 Dimensional Rosenzweig–MacArthur equations

Combining logistic growth, total prey consumption $g(H)P$, conversion and mortality gives the project's Rosenzweig–MacArthur model [17]:

$$\frac{dH}{dt} = rH\left(1 - \frac{H}{K}\right) - \frac{aHP}{b+H}, \tag{11}$$

$$\frac{dP}{dt} = \frac{eaHP}{b+H} - mP. \tag{12}$$

These are exactly the dimensional equations in the shared repository [12]. Logistic growth and saturation allow stability to depend on parameters. Whether a specified case approaches an equilibrium or exhibits a stable cycle requires the subsequent analysis and numerical evidence; it does not follow merely from naming the model.

3 Units and assumptions

Let U_H and U_P denote the chosen hare and lynx index units. Keeping these labels distinct prevents an unjustified interpretation of conversion efficiency.

Symbol	Meaning	Units
t	Elapsed time	year
H, P	Hare and lynx states	U_H, U_P
r	Low-density hare net growth rate	year^{-1}
K	Effective prey-only carrying-capacity index	U_H
a	Effective saturated consumption coefficient	$U_H U_P^{-1}$ year^{-1}
b	Prey index at half saturation	U_H
e	Effective predator conversion coefficient	$U_P U_H^{-1}$
m	Predator mortality rate	year^{-1}
H_0, P_0	States at the initial time	U_H, U_P

All six parameters are strictly positive. For example, $[aHP/(b+H)] = U_H/\text{year}$ and $[eaHP/(b+H)] = U_P/\text{year}$, matching the two left-hand sides. In separately scaled abundance indices, e is an effective conversion factor. It need not be a dimensionless biomass efficiency bounded above by one. For this historical iteration U_H means thousands of hare pelts and U_P thousands of lynx pelts. The rates r, m retain their growth/mortality interpretation only conditional on O1 and an adequate dynamical model; K, b, a, e are effective pelt-unit coefficients. Scale invariance does not establish accurate recovery of any biological parameter.

The simplifying assumptions and their consequences are:

1. A closed two-species system has no immigration, emigration, alternative prey or other predators. This isolates the proposed feedback but omits a food web.

2. Continuous, spatially mixed states omit age structure, territories and local differences. The approximation is unsuitable for individual-level extinction claims.

3. Constant parameters omit seasons, weather and changing resource availability. Fitted rates summarize the chosen interval rather than universal ecological constants.

4. Resource limitation is represented by logistic prey growth. Plants are not modelled as a third dynamical state.

5. Predation follows the type II response with no interference between predators. Predator growth depends on prey; without prey, P decays at rate m.

6. Pelt totals are treated as noisy abundance proxies. Constant observation scales are a working assumption. Unknown or changing capture rates can confound parameter inference. Harvest is not modelled as an additional removal process.

3.1 Basic consistency checks

The equations have the form $\dot{H} = HA(H,P)$, $\dot{P} = PB(H)$ with continuous coefficients for $H \geq 0$. Along a finite solution interval,

$$H(t) = H_0 \exp\left(\int_0^t A(H(s),P(s))\,ds\right), \quad P(t) = P_0 \exp\left(\int_0^t B(H(s))\,ds\right). \tag{13}$$

Positive initial states remain positive, and either zero boundary is invariant by uniqueness. A numerical method can still introduce negative approximations, so its outputs must be checked. Also $\dot{H} \leq rH(1 - H/K)$, giving $H(t) \leq \max\{H_0, K\}$ for $t \geq 0$. Since $\dot{P} \leq eaP$, predator abundance cannot blow up in finite time. These checks establish basic consistency, not ecological accuracy or long-term stability.

4 Non dimensionalisation

4.1 Choosing the scales

For a fixed parameter vector define

$$s = rt, \qquad x = H/K, \qquad y = \frac{aP}{rK}. \tag{14}$$

The corresponding reference time and states are $1/r$, K and rK/a. Their units are year, U_H and U_P. The scales therefore make all three new variables dimensionless. Choosing K for prey makes the prey-only capacity one; choosing rK/a for predators makes the prey-loss coefficient one. This is the same notation used in Joel's existing dimensionless analysis [10].

Define the three dimensionless groups

$$\beta = b/K, \qquad \alpha = ea/r, \qquad \delta = m/r. \tag{15}$$

They compare half saturation with carrying capacity, maximum predator growth with prey growth, and predator mortality with prey growth, respectively. This scaling also means a sweep of K changes both β and the state scales. If the dimensional initial states are held fixed in that sweep, their scaled values must be recomputed for each K.

4.2 Substitution and chain rule

Because $H = Kx$, $P = (rK/a)y$ and $ds/dt = r$,

$$\frac{dH}{dt} = rK\frac{dx}{ds}, \qquad \frac{dP}{dt} = \frac{r^2K}{a}\frac{dy}{ds}. \tag{16}$$

Substitution into the prey equation gives

$$rK\frac{dx}{ds} = rKx(1-x) - \frac{rKxy}{b/K+x}. \tag{17}$$

Substitution into the predator equation gives

$$\frac{r^2K}{a}\frac{dy}{ds} = \frac{erKxy}{b/K+x} - \frac{mrK}{a}y. \tag{18}$$

Dividing by the respective factors yields

$$\frac{dx}{ds} = x(1-x) - \frac{xy}{\beta+x}, \tag{19}$$

$$\frac{dy}{ds} = \alpha\frac{xy}{\beta+x} - \delta y. \tag{20}$$

The two systems describe the same trajectories after the changes of units. Reducing six dimensional parameters to three groups does not prove that all six can be identified from data: physical scales and initial conditions also carry information.

4.3 Initial conditions and conversion back

At year 1900, $s = 0$, $x_0 = H_0/K$ and $y_0 = aP_0/(rK)$. Recover the dimensional variables using

$$\text{Year} = 1900 + s/r, \qquad H = Kx, \qquad P = (rK/a)y. \tag{21}$$

If r, K, a are separately specified, recover $b = \beta K$, $e = \alpha r/a$ and $m = \delta r$. The three dimensionless groups alone cannot reconstruct all six dimensional parameters.

For an arithmetic check, use the repository's synthetic values [12]:

$$(r, K, a, b, e, m) = (1, 60, 2, 20, 0.6, 0.5), \qquad (H_0, P_0) = (35, 9).$$

These give

$$(\beta, \alpha, \delta) = (1/3, 1.2, 0.5), \qquad (x_0, y_0) = (7/12, 0.3). \tag{22}$$

At this initial state, $(\dot{H}, \dot{P}) = (413/132, 261/110)$ in their respective index units per year. The dimensionless derivatives are $(dx/ds, dy/ds) = (413/7920, 87/1100)$; multiplying by $rK = 60$ and $r^2K/a = 30$ recovers the dimensional values. These are synthetic checks, not fitted historical estimates.

An equally valid alternative is $X = H/b = x/\beta$ and $Y = aP/(rb) = y/\beta$, with the same time s. Writing $\kappa = K/b = 1/\beta$ gives $X_s = X(1-X/\kappa) - XY/(1+X)$ and $Y_s = \alpha XY/(1+X) - \delta Y$. This keeps state scales fixed during a K sweep when other parameters remain fixed. We use the K-based form as the main convention to match the existing team analysis, while retaining the alternative as an explanation of the scaling tradeoff.

5 Compatibility with data scaling and the PINN

5.1 Keep mathematical and numerical coordinates explicit

The historical PINN [21] uses fixed training maxima s_H, s_P, normalized outputs $u = H/s_H$, $v = P/s_P$, training duration $T = 15$ years, and internal time input $\xi = 2t/T - 1$. It differentiates with respect to elapsed years t. These numerical choices can coexist with the dimensionless theory coordinates. In elapsed time,

$$\frac{du}{dt} = ru(1 - s_Hu/K) - \frac{as_Puv}{b + s_Hu}, \tag{23}$$

$$\frac{dv}{dt} = \frac{eas_Huv}{b + s_Hu} - mv. \tag{24}$$

The bridge to the theoretical variables is

$$x = (s_H/K)u, \qquad y = \frac{as_P}{rK}v, \qquad s = \frac{rT}{2}(\xi + 1). \tag{25}$$

If differentiation is instead performed with respect to ξ, the chain rule requires $du/dt = (2/T)du/d\xi$, and likewise for v. Thus a residual formed in years is $R_u = (2/T)u_\xi - f_H/s_H$. If one forms a residual in ξ coordinates, $\widetilde{R}_u = u_\xi - (T/2)f_H/s_H = (T/2)R_u$. With the same averaging convention, its squared penalty is $(T/2)^2$ larger; the loss coefficient must be multiplied by $(2/T)^2$ to preserve the same objective. This is a unit change, not evidence that one trained model is better.

Because all six coefficients, including r and a, are inferred in this iteration, directly using rt and $aP/(rK)$ as fixed training inputs or targets would introduce parameter-dependent coordinates. Retaining his fixed training normalization avoids that complication. The dimensionless form is for derivation, interpretation and consistent mapping, and does not require rewriting the PINN.

5.2 Why measurement units change parameter meanings

For any constant positive changes $\widetilde{H} = c_H H$, $\widetilde{P} = c_P P$, keeping time in years preserves the model form if

$$\widetilde{r} = r, \quad \widetilde{m} = m, \quad \widetilde{K} = c_H K, \quad \widetilde{b} = c_H b, \quad \widetilde{a} = \frac{c_H}{c_P}a, \quad \widetilde{e} = \frac{c_P}{c_H}e. \tag{26}$$

This follows by substituting into both derivatives, including the denominator. It leaves $\beta, \alpha, \delta, x, y$ unchanged. For example, converting both species from counts to thousands uses $c_H = c_P = 10^{-3}$: K and b divide by 1000, while a and e keep their numerical values. Unequal species scales change a and e too. Unknown pelt-to-population multipliers have the same algebraic effect, but their unknown values prevent conversion to absolute biological rates. Time-varying multipliers would add derivative terms and are outside this model.

The same transformation leaves Joel's stability ratio $\sigma = K/K_H$ unchanged, where $K_H = b(ea + m)/(ea - m)$ and positive coexistence is applicable [10]. Thus the dimensionless groups have a model-based interpretation independent of constant observation scaling. This does not confirm that the real ecosystem follows the model or has a stable limit cycle.

With the observed predator scale fixed, an arbitrary rescaling of P cannot preserve the same observations. Moreover, a occurs separately in the prey equation while ea occurs in the predator equation. Therefore the scale ambiguity alone does not prove that effective a and e are inseparable. It also does not prove that all six coefficients are well determined by 16 annual pairs.

5.3 Reference settings for the completed iteration

The reference is `configs/historical.json`, rather than the earlier synthetic configuration [21]. State order is (H, P) and parameter names are (r, K, a, b, e, m), matching the shared right-hand side. All six coefficients are fitted. Their independent positive bounds are

$$r, m \in [0.05, 3], \quad K \in [20, 500], \quad a \in [0.05, 10], \quad b \in [0.5, 300], \quad e \in [0.02, 5].$$

The units are those in the table above. These are exploratory numerical bounds, not measured ecological ranges. The completed iteration imposes neither $e \leq 1$ nor a coexistence constraint. This describes the recorded computation, rather than a new assertion of identifiability or a proposed change to the inverse problem.

Historical fitting and ODE forecasts use `model.solve`: RK45 with relative tolerance 10^{-8}, absolute tolerance 10^{-10} and maximum step 0.1 years. Rishi's tighter settings for illustrative

long-run simulations are a separate calculation, not the settings of this historical comparison [21, 15].

In the main analysis the ODE starts at the observed 1900 index state $(30, 4)$. The PINN enforces its initial state with a soft penalty and reports the discrepancy; we do not describe this as an exact network constraint. In the fitted-initial-state sensitivity, each method forecasts from its fitted state at 1900. Every forecast integrates from 1900 through 1920 with frozen parameters; none restarts from a test-year observation. The historical handoff records both sensitivity variants and preserves them alongside the main result [21].

6 Validation responsibilities and limits

The derivations supply a model to test, not a claim that a published model is novel or that a PINN earns bonus marks automatically. The independent contribution here is the explicit reasoning, unit audit, coordinate transformations and interpretation under the chosen observation convention. The accompanying notebook checks algebraic equivalence at multiple exact rational inputs. These checks do not replace empirical validation or establish parameter identifiability.

Joel's contribution supplies equilibrium, stability and Hopf analysis [10]; Rishi's supplies numerical checks and parameter-regime illustrations [15]. Advay's data records and Yash's historical fits, matching least-squares benchmark and evaluation are available [1, 21]. Their empirical results belong in the corresponding report sections. Comparisons use the same training years, parameter bounds, initial-state convention and solver settings, with separate hare and lynx MAE and RMSE in thousands of pelts. The PINN's soft initial penalty and its network–ODE discrepancy must remain visible qualifications.

Five held-out years offer limited evidence. Synthetic recovery generated by the same equations checks the inverse procedure under favourable assumptions; it is not independent evidence that those equations describe real ecology. A small training error does not show unique biological parameters. Any substantive change after the completed evaluation must be documented as a later revision, preserving the existing results. Report claims must remain restricted to the mathematical and empirical evidence actually obtained.

7 Equilibria, Stability and Hopf Bifurcation

Throughout this section we analyse the dimensional Rosenzweig–MacArthur model

$$\frac{dH}{dt} = f(H,P) = rH\Big(1-\frac{H}{K}\Big) - \frac{aHP}{b+H}, \qquad \frac{dP}{dt} = g(H,P) = \frac{eaHP}{b+H} - mP, \tag{27}$$

with all six parameters r, K, a, b, e, m strictly positive, as defined in the model section. Every stability statement below is tied either to a derivation given here, to a cited theorem, or to a numerical check; Table 3 at the end of the section records which is which.

7.1 Well-posedness: invariance and boundedness

Lemma 1. *The closed quadrant $\{H \geq 0,\, P \geq 0\}$, both coordinate axes, and the open quadrant are invariant under* (27)*. Every solution starting in the quadrant is bounded, with*

$$\limsup_{t\to\infty} H(t) \leq K, \qquad \limsup_{t\to\infty}\big(eH(t)+P(t)\big) \leq \frac{eK(r+m)^2}{4rm}.$$

Proof. Both right-hand sides factor as $H\cdot(\cdot)$ and $P\cdot(\cdot)$, so $H=0$ and $P=0$ are invariant lines and, by uniqueness of solutions, no orbit can cross them. Since $\dot H \leq rH(1-H/K)$, comparison with the logistic equation gives $\limsup H \leq K$. For $W = eH + P$ the predation terms cancel: $\dot W = erH(1-H/K) - mP = eH\big(r(1-H/K)+m\big) - mW$. The quadratic $H\big(r(1-H/K)+m\big)$ is maximised at $H = K(r+m)/(2r)$ with value $K(r+m)^2/(4r)$, so $\dot W \leq eK(r+m)^2/(4r) - mW$ and the bound follows. □

Boundedness is what later allows the Poincaré–Bendixson theorem to be used.

7.2 Equilibria and biological feasibility

Setting $f = g = 0$:

- $P = 0$ in $g = 0$ leaves $rH(1-H/K) = 0$, giving the *origin* $E_0 = (0,0)$ and the *prey-only* equilibrium $E_1 = (K, 0)$.
- $P \neq 0$ requires $eaH/(b+H) = m$, i.e.

$$H^* = \frac{mb}{ea-m}, \tag{28}$$

and then $f = 0$ with $H = H^* \neq 0$ gives

$$P^* = \frac{r}{a}\Big(1-\frac{H^*}{K}\Big)(b+H^*) = \frac{ebr\big(K(ea-m)-bm\big)}{K(ea-m)^2}. \tag{29}$$

This is the *coexistence* equilibrium $E^* = (H^*, P^*)$.

Proposition 1 (Feasibility)**.** *E_0 and E_1 always lie in the closed quadrant. E^* lies in the open quadrant if and only if*

$$ea > m \quad \text{and} \quad H^* < K \qquad \Longleftrightarrow \qquad m < \frac{eaK}{b+K}.$$

Proof. $H^* > 0$ iff $ea > m$. Given this, $P^* > 0$ iff $1 - H^*/K > 0$. Rearranging $mb/(ea-m) < K$ gives $m(b+K) < eaK$. □

Biologically, ea is the predator's maximal per-capita birth rate (attained when prey is abundant), so $ea > m$ says lynx can outgrow their mortality at all; $H^* < K$ says the hare level lynx need to break even is actually reachable by the prey on its own. Note that H^* does not depend on r or K: the prey level at coexistence is set entirely by the predator.

7.3 Jacobian

Differentiating (27),

$$J(H,P) = \begin{pmatrix} r\Big(1-\dfrac{2H}{K}\Big) - \dfrac{abP}{(b+H)^2} & -\dfrac{aH}{b+H} \\ \dfrac{eabP}{(b+H)^2} & \dfrac{eaH}{b+H} - m \end{pmatrix}. \tag{30}$$

For a planar system, with $\tau = \operatorname{tr} J$ and $\Delta = \det J$, an equilibrium is a saddle if $\Delta < 0$; is asymptotically stable if $\Delta > 0,\ \tau < 0$; and is unstable if $\Delta > 0,\ \tau > 0$. It is a focus when $\tau^2 < 4\Delta$ and a node when $\tau^2 > 4\Delta$.

7.4 Local stability of each equilibrium

Origin. $J(E_0) = \operatorname{diag}(r, -m)$, with eigenvalues $r > 0$ and $-m < 0$. E_0 is *always a saddle*: the P-axis is its stable manifold (lynx alone die out) and the H-axis its unstable manifold (hares alone grow). Joint extinction is never an attractor.

Prey-only equilibrium.

$$J(E_1) = \begin{pmatrix} -r & -\dfrac{aK}{b+K} \\ 0 & \dfrac{eaK}{b+K} - m \end{pmatrix}, \qquad \lambda_1 = -r, \quad \lambda_2 = \frac{eaK}{b+K} - m.$$

E_1 is a stable node when $\lambda_2 < 0$, i.e. exactly when E^* is *not* feasible, and a saddle when $\lambda_2 > 0$, i.e. exactly when E^* is feasible. At $K = H^*$ the two equilibria coincide and exchange stability (a transcritical bifurcation). λ_2 is the invasion rate of a rare lynx population into a hare population at carrying capacity.

Coexistence equilibrium. At E^* the predator equation gives $eaH^*/(b+H^*) = m$, so $J_{22} = 0$. Using the prey nullcline, $aP^*/(b+H^*) = r(1-H^*/K)$, to eliminate P^* from J_{11}:

$$J_{11} = r\Big(1-\frac{2H^*}{K}\Big) - \frac{b}{b+H^*}\, r\Big(1-\frac{H^*}{K}\Big) = \frac{rH^*\,(K-b-2H^*)}{K\,(b+H^*)}.$$

Hence

$$\tau(E^*) = \frac{rH^*}{K(b+H^*)}\,\big(K - K_H\big), \qquad K_H := b + 2H^* = \frac{b\,(ea+m)}{ea-m}, \tag{31}$$

$$\Delta(E^*) = -J_{12}J_{21} = \frac{aH^*}{b+H^*} \cdot \frac{eabP^*}{(b+H^*)^2} = \frac{rmb\,(K-H^*)}{K(b+H^*)}. \tag{32}$$

Theorem 1 (Local stability of E^*)**.** *Suppose E^* is feasible ($H^* < K$). Then $\Delta(E^*) > 0$, so E^* is never a saddle, and*

E^* is locally asymptotically stable if $K < K_H$, $\qquad$ E^* is unstable (a repeller) if $K > K_H$.

Proof. $\Delta > 0$ follows from (32) since $K > H^*$. The sign of τ is the sign of $K - K_H$ by (31). Apply the trace–determinant classification. □

Remark 1. The intrinsic hare growth rate r appears in τ and Δ only as a positive prefactor. It therefore sets how fast trajectories spiral, but has *no effect* on whether E^* is stable. Near feasibility (K just above H^*), $\Delta \to 0$ while $\tau < 0$, so E^* is a stable node; as K increases it becomes a stable focus once $\tau^2 < 4\Delta$ (at $K \approx 42.5$ for the illustrative parameters of §7.8), so convergence is oscillatory well before the Hopf point.

Table 1: Equilibria of (27) and their local stability.

Equilibrium	Feasible when	Eigenvalues / invariants	Type
$E_0 = (0, 0)$	always	$r,\ -m$	saddle
$E_1 = (K, 0)$	always	$-r,\ \frac{eaK}{b+K} - m$	stable node if $K < H^*$ or $ea \le m$; saddle if $K > H^*$
$E^* = (H^*, P^*)$	$ea > m,\ H^* < K$	$\tau \propto K - K_H,\ \Delta > 0$	stable if $K < K_H$; unstable if $K > K_H$

7.5 Nullclines and the phase plane

Prey nullclines ($\dot{H} = 0$): the P-axis $H = 0$, and the parabola

$$P = \phi(H) := \frac{r}{a}\Big(1 - \frac{H}{K}\Big)(b + H), \tag{33}$$

which is concave, vanishes at $H = -b$ and $H = K$, cuts the P-axis at rb/a, and peaks at

$$H_{\text{peak}} = \frac{K - b}{2}, \qquad \phi(H_{\text{peak}}) = \frac{r(K + b)^2}{4aK}.$$

$\dot{H} > 0$ below the parabola and $\dot{H} < 0$ above it (for $H > 0$).

Predator nullclines ($\dot{P} = 0$): the H-axis $P = 0$, and the vertical line $H = H^*$. $\dot{P} > 0$ to the right of $H = H^*$ (enough prey) and $\dot{P} < 0$ to the left.

Flow. The two interior nullclines divide the open quadrant into four regions. Starting below the parabola to the right of H^*, both species increase; crossing the parabola hares start to fall while lynx still rise; crossing $H = H^*$ both fall; crossing back below the parabola hares recover while lynx are still falling. The flow therefore circulates *counter-clockwise* around E^* in the (H, P) plane, and lynx peaks lag hare peaks. Near the Hopf point the lag is a quarter-period; it grows with K (numerically, 2.5 to 4.5 years for regimes D–F below).

Graphical stability criterion. Since $K_H = b+2H^*$ is equivalent to $H^* = (K-b)/2 = H_{\text{peak}}$, Theorem 1 says:

E^* is stable if the predator nullcline cuts the prey parabola to the right of its hump, and unstable if it cuts to the left.

This is the original Rosenzweig–MacArthur graphical criterion [18]. It also gives the sign of J_{11} directly: $J_{11} = a\,H^*\phi'(H^*)/(b + H^*)$, so a rising nullcline (left of the hump) means local prey self-amplification ($J_{11} > 0$). Figure 1 shows both cases.

7.6 Hopf bifurcation with K as the parameter

We fix r, a, b, e, m with $ea > m$ and vary K over (H^*, ∞). Note H^* is independent of K, while P^* increases with K.

Theorem 2 (Hopf bifurcation)**.** *At $K = K_H = b + 2H^*$ the coexistence equilibrium undergoes a Hopf bifurcation:*

(i) $\tau(E^*) = 0$ and $\Delta(E^*) = \dfrac{rmb\,(K_H - H^*)}{K_H(b + H^*)} = \dfrac{rmb}{K_H} > 0$, so $J(E^*)$ has eigenvalues $\pm i\omega_0$ with

$$\omega_0 = \sqrt{\frac{rmb}{b + 2H^*}}, \qquad T_0 = \frac{2\pi}{\omega_0};$$

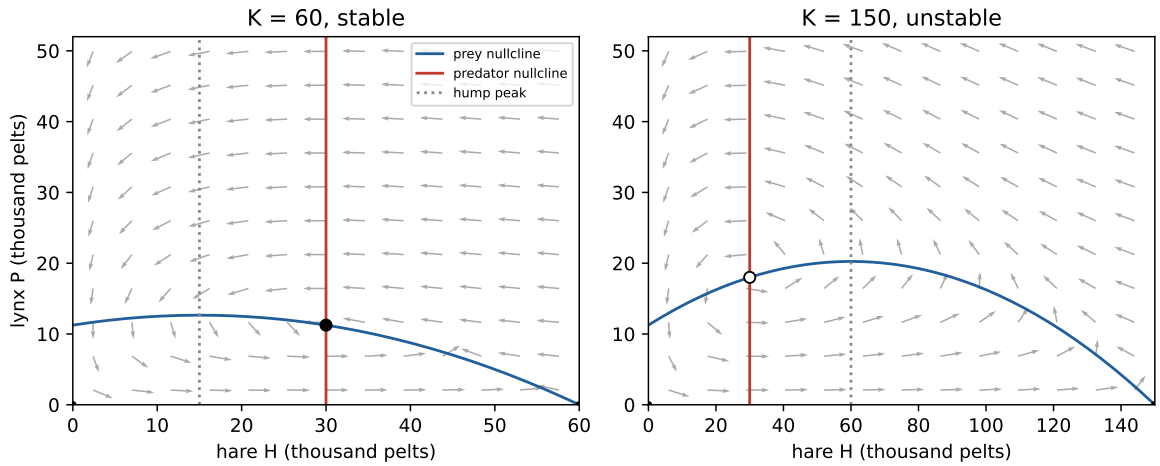


Figure 1: Nullclines and direction field for the illustrative parameters ($r = 1.5$, $a = 4$, $b = 30$, $e = 0.4$, $m = 0.8$; $H^* = 30$, $K_H = 90$). Left: $K = 60$; the predator nullcline meets the parabola right of its hump and E^* (filled) is a stable focus. Right: $K = 150$; the intersection is left of the hump and E^* (open) is unstable. Small dots mark E_0 and E_1.

(ii) (transversality) the real part of the eigenvalues, $\tau/2$, crosses zero with nonzero speed:

$$\frac{d}{dK}\left(\frac{\tau}{2}\right)\Big|_{K=K_H} = \frac{rH^*}{2(b+H^*)(b+2H^*)} > 0.$$

Proof. (i) Substitute $K = K_H$ into (31) and (32); $K_H - H^* = b + H^*$ simplifies Δ. (ii) Differentiate (31) in K: $\frac{d}{dK}\frac{K-K_H}{K} = \frac{K_H}{K^2}$, which equals $1/K_H$ at $K = K_H$. All symbolic steps were checked with SymPy (`scripts/joel_symbolic_derivation.py`) and the closed forms are tested against the shared model code (`tests/test_stability.py`). □

Conditions (i)–(ii) establish that a pair of complex eigenvalues crosses the imaginary axis. To conclude that a small periodic orbit is born, and on which side, the Hopf theorem also needs the first Lyapunov coefficient ℓ_1 to be nonzero [11]: $\ell_1 < 0$ gives a *supercritical* bifurcation (a stable cycle for $K > K_H$) and $\ell_1 > 0$ a subcritical one.

What is analytic and what is numerical. We separate four statements.

1. **Analytic (derived here):** the threshold K_H, conditions (i)–(ii), and the exchange of stability at E^*.

2. **Analytic (derived here): existence of a periodic orbit for $K > K_H$.** Take any initial condition in the open quadrant other than E^*. By Lemma 1 its ω-limit set Ω is nonempty and compact. Ω cannot contain E^*, which is a repeller. Ω cannot be $\{E_1\}$, because the stable manifold of the saddle E_1 is the H-axis. If Ω contained E_1 together with other points, it would contain a point of E_1's stable manifold on the H-axis and hence the whole axis orbit through it, which tends backward to E_0; repeating the argument at E_0 would put a point of the P-axis in Ω, whose backward orbit is unbounded, contradicting compactness. So Ω contains no equilibria, and the Poincaré–Bendixson theorem [20] makes Ω a periodic orbit.

3. **Cited theorems:** Cheng [4] proved the limit cycle is *unique* for $K > K_H$; since every interior orbit other than E^* tends to a periodic orbit, that unique cycle attracts them all and is orbitally stable. Hsu [9] proved that a locally stable E^* is *globally* stable in the open quadrant whenever the prey nullcline is concave, which our parabola (33) is, so E^*

attracts every interior orbit for $H^* < K < K_H$. We use these results without reproducing their proofs.

4. **Numerical (for our parameter set only):** evaluating Kuznetsov's formula for ℓ_1 at $K = K_H$ gives $\ell_1 \approx -4.0 \times 10^{-4} < 0$ (and -1.3×10^{-3} for a second set $r = 0.7$, $m = 0.5$, $b = 20$), so the bifurcation is supercritical, consistent with the cited uniqueness result. Simulations confirm the supercritical signature: the post-transient cycle half-amplitude in H scales as $(K - K_H)^{1/2}$ (fitted log–log slope 0.507 for $K - K_H \in [0.5, 8]$), and the period tends to T_0.

None of this says that the *real* hare–lynx data sit on a limit cycle. That depends on whether fitted parameters satisfy $K > K_H$, and with about two oscillations in 1900–1920 a slowly damped oscillation (K slightly below K_H) can fit the same data. That question belongs to the fitting sections and must be answered there with uncertainty.

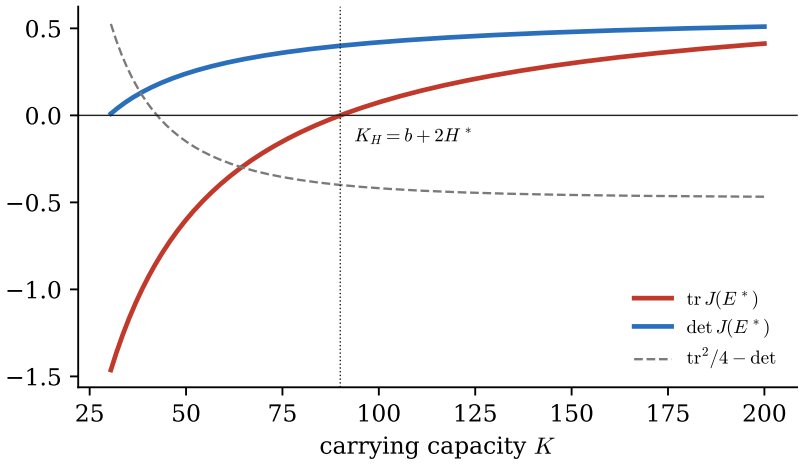


Figure 2: Trace and determinant of $J(E^*)$ as functions of K (illustrative parameters). The determinant stays positive for all feasible $K > H^* = 30$, while the trace changes sign at $K_H = 90$. The dashed curve is negative (focus) for $K > 42.5$.

7.7 The paradox of enrichment

Increasing K (enriching the hares' environment) leaves H^* unchanged and only raises P^*: every extra unit of carrying capacity is passed to the predator. Graphically, the prey parabola's hump $(K - b)/2$ moves right while the predator nullcline stays at H^*, so for large enough K the intersection falls left of the hump and the equilibrium destabilises. Past K_H the system settles on a limit cycle whose amplitude grows with K. For our illustrative parameters, at $K = 150$ the hare minimum on the cycle is about 1.1 thousand, compared with $H^* = 30$. In a finite, stochastic population such deep troughs bring a real risk of extinction. Enrichment thus *destabilises* the system, the paradox named by Rosenzweig [16].

The mechanism is the saturating Holling type II response. At low H each lynx's consumption rises almost linearly with prey, which damps fluctuations; near and above b it saturates, so per-capita predation risk $aP/(b+H)$ *falls* as hares increase. That positive feedback on prey growth is exactly the term making $J_{11} > 0$ left of the hump.

Link to Lotka–Volterra. In the classical model ($K = \infty$, linear response) the analogous Jacobian at coexistence has $\tau \equiv 0$ for every parameter value, so it sits permanently on the Hopf boundary. That is why its neutral cycles are structurally unstable: any small change in the

model's form moves it to one side or the other. Rosenzweig–MacArthur makes the side explicit through the sign of $K - K_H$.

7.8 Illustrative regimes and numerical validation targets

These parameters are *illustrative, not fitted*: H and P in thousands of pelts, t in years, with $r = 1.5,\ a = 4,\ b = 30,\ e = 0.4,\ m = 0.8$. Then $ea = 1.6 = 2m$, so $H^* = b = 30$, $K_H = 90$, $\omega_0 = \sqrt{0.4} \approx 0.632$ and $T_0 \approx 9.94$ years, close to the observed decadal cycle. Table 2 gives the expected behaviour on each side of the threshold; the simulation section reproduces these with RK45 (`rtol`$= 10^{-10}$, `atol`$= 10^{-12}$, 3000-year burn-in, start $(20, 5)$).

Table 2: Regimes in K (other parameters fixed). Eigenvalues are of $J(E^*)$, except in A where they are of $J(E_1)$.

	K	Linear prediction	Eigenvalues	Long-run behaviour (simulated)
A	25	E^* infeasible ($K < H^*$)	$-1.5,\ -0.073$	lynx die out; $\to (25, 0)$
B	60	E^* stable focus	$-0.188 \pm 0.515i$	$\to$ (30, 11.25), e-fold ≈ 5 yr
C	85	just below K_H	$-0.022 \pm 0.623i$	$\to$ (30, 14.56), e-fold ≈ 45 yr
D	90	Hopf point	$\pm 0.632i$	very slow (algebraic) decay
E	95	just above K_H	$+0.020 \pm 0.640i$	cycle: $H \in [15.8, 50.9]$, $P \in [9.8, 20.8]$, $T = 10.25$
F	150	well above K_H	$+0.150 \pm 0.676i$	cycle: $H \in [1.1, 134.3]$, $P \in [1.7, 41.2]$, $T = 13.91$

Two cautions for the numerics. At D the equilibrium is a weak focus ($\ell_1 < 0$), so oscillations decay only like $t^{-1/2}$; after 3000 years the residual amplitude in H is still about 1.5, which is expected and is not a sign of a cycle. The cycle at E is already large because the amplitude coefficient is large ($A_H \approx 7.8\sqrt{K - K_H}$), so "small amplitude" only holds within about one unit of K_H.

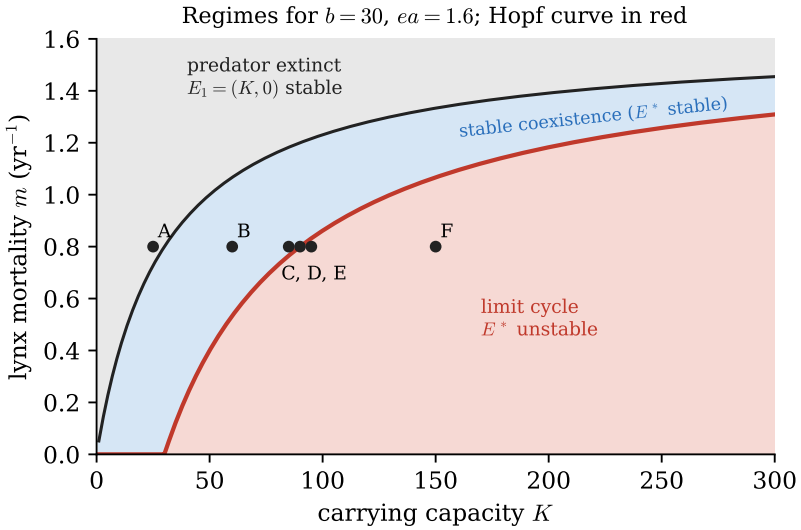


Figure 3: Analytic regimes in the (K, m) plane for $b = 30$, $ea = 1.6$. Black curve: $m = eaK/(b + K)$ (predator persistence boundary). Red curve: $m = ea(K - b)/(K + b)$ (Hopf curve). Dots mark regimes A–F of Table 2.

7.9 Sensitivity of the threshold

Because $K_H = b(ea+m)/(ea-m)$ is explicit, sensitivity can be read off exactly. The elasticities (percentage change in K_H per percentage change in a parameter) are

$$\frac{\partial \ln K_H}{\partial \ln b} = 1, \qquad \frac{\partial \ln K_H}{\partial \ln m} = -\frac{\partial \ln K_H}{\partial \ln a} = -\frac{\partial \ln K_H}{\partial \ln e} = \frac{2\,ea\,m}{(ea)^2 - m^2}, \qquad \frac{\partial \ln K_H}{\partial \ln r} = 0.$$

At the illustrative values ($ea = 2m$) the middle elasticity is 4/3: a 10% rise in lynx mortality raises the threshold by about 13%, and so stabilises the system. The elasticity blows up as $m \to ea$, where the predator is barely viable and $K_H \to \infty$. Weak predators stabilise.

Equivalently, for fixed K the regimes are bands in m (Figure 3):

$$\underbrace{m < \frac{ea(K-b)}{K+b}}_{\text{limit cycle}} \qquad \underbrace{\frac{ea(K-b)}{K+b} < m < \frac{eaK}{K+b}}_{\text{stable coexistence}} \qquad \underbrace{m > \frac{eaK}{K+b}}_{\text{lynx extinct}}$$

A useful single number for any fitted parameter set is the *stability margin* $\sigma = K/K_H$, when coexistence is feasible: $\sigma < 1$ gives a locally stable equilibrium (node or focus), while $\sigma > 1$ gives an unstable equilibrium and the model's attracting cycle. Reporting σ (with its spread over fits) is the cleanest way to say whether the data support sustained cycling.

For inference, a and e enter H^* and K_H through ea. If an additional unknown predator observation scale is fitted, the transformation $(P, a, e) \mapsto (cP, a/c, ec)$ introduces a scale ambiguity. In this historical iteration, the observed pelt-state units are fixed and no scale factor is fitted, so this transformation changes the observed predator values and does not by itself prove non-identifiability of the effective coefficients. All six are fitted. Their practical uncertainty is assessed using training diagnostics and the declared sensitivity runs, and a, e are not interpreted as uniquely recovered absolute biological rates.

7.10 Dimensionless form of the results

With the scaling $x = H/K$, $y = aP/(rK)$, $s = rt$ (to be reconciled with the frozen scaling in the model section), (27) becomes

$$x' = x(1-x) - \frac{xy}{\beta + x}, \qquad y' = y\Big(\frac{\alpha x}{\beta + x} - \delta\Big), \qquad \beta = \frac{b}{K},\ \alpha = \frac{ea}{r},\ \delta = \frac{m}{r}.$$

Then $x^* = \beta\delta/(\alpha - \delta)$, coexistence requires $\alpha > \delta$ and $x^* < 1$, and the Hopf condition $K = K_H$ reads $\beta(\alpha+\delta)/(\alpha-\delta) = 1$, i.e. $\beta + 2x^* = 1$. Only three combinations of the six parameters control the qualitative dynamics.

Table 3: Status of each stability claim in this section.

Claim	Evidence
Equilibria, feasibility, local stability of E_0, E_1, E^*	Derived here; SymPy-checked
Hopf threshold $K_H = b + 2H^*$, $\Delta > 0$, transversality	Derived here; SymPy-checked
A periodic orbit exists for every $K > K_H$	Derived here (Poincaré–Bendixson)
Cycle is unique and attracting for $K > K_H$	Cited theorem [4]
E^* globally stable for $H^* < K < K_H$	Cited theorem [9]
Supercritical Hopf ($\ell_1 < 0$)	Numerical evaluation of analytic formula, two parameter sets
$\sqrt{K - K_H}$ amplitude law; periods, amplitudes, lags	Numerical (RK45), illustrative parameters
Real hare–lynx data lie on a limit cycle	*Not established here*; requires $\sigma > 1$ at fitted values with uncertainty

8 Numerical Simulations

Section 7 locates the Hopf threshold $K_H = b + 2H^*$ analytically. Linearisation is, however, a local statement about the equilibrium at a single parameter value. This section tests those predictions numerically and measures what linear analysis cannot give: the long-run behaviour away from E^*, the size and period of the cycle, and the shape of the bifurcation. Every trajectory is computed with the group's shared right-hand side of (27).

8.1 Method

Parameters. We use the illustrative, unfitted values of Section 7: $r = 1.5$, $a = 4$, $b = 30$, $e = 0.4$, $m = 0.8$ (H, P in thousands of pelts, t in years), for which $H^* = 30$, $K_H = 90$ and $T_0 = 2\pi/\omega_0 \approx 9.94$ years. The carrying capacity K is varied.

Solver. Equations (27) are integrated with the adaptive Dormand–Prince Runge–Kutta pair (RK45) in `scipy.integrate.solve_ivp` [5, 19]. At each step a fourth- and a fifth-order estimate are formed from the same seven slope evaluations; their difference estimates the local error, and the step is rejected and reduced whenever this exceeds $\texttt{rtol}\,|y| + \texttt{atol}$. The project uses two settings of this solver for two different jobs. Fitting and the 1900–1920 forecasts of the historical sections use `model.solve` with $\texttt{rtol} = 10^{-8}$, $\texttt{atol} = 10^{-10}$ and a maximum step of 0.1 years, identical for both fitting methods. The long-run simulations in this section run for thousands of years, so they use tighter tolerances, $\texttt{rtol} = 10^{-10}$ and $\texttt{atol} = 10^{-12}$, without a step cap. Both call the same shared right-hand side. The illustrative runs start from $(H_0, P_0) = (20, 5)$; all values are in `configs/simulations.json`.

Long-run behaviour. To describe the attractor we integrate for a burn-in period T_{burn}, discard it, and record the next 300–400 years. On the recorded window we report the range of H and P; if H still oscillates, the period is the mean interval between successive hare maxima and the lag is the mean delay from a hare maximum to the next lynx maximum. We take $T_{\text{burn}} = 3000$ years, increased to 20,000 years when $|K - K_H| < 1$, where the equilibrium is a weak focus and transients decay very slowly.

8.2 Solver accuracy

The classical Lotka–Volterra system $\dot{H} = \alpha H - \beta HP$, $\dot{P} = \delta HP - \gamma P$ conserves $V = \delta H - \gamma \ln H + \beta P - \alpha \ln P$, so any drift in V along a computed orbit is numerical error. Over 100 time units RK45 keeps $|V(t) - V(0)|$ below 4×10^{-10}, whereas explicit Euler with step $h = 0.01$ drifts by 0.15 and its orbit spirals outward (Figure 4), creating oscillatory growth the model does not have. For Rosenzweig–MacArthur at $K = 150$, tightening both tolerances a hundredfold changes the solution by a relative 8×10^{-10}, and all states remain positive, as Lemma 1 requires.

8.3 Reproduction of the analytic regimes

Table 4 reproduces each validation target of Section 7 (3000-year burn-in throughout). Regime D lies exactly on K_H: the residual oscillation there is a slowly decaying transient (decay is only algebraic at a weak focus), not a cycle.

Figure 5 shows four of these regimes. Below the threshold (B, C) the predator nullcline meets the prey hump to the right of its peak and trajectories spiral into E^*; at $K = 85$ the trace is only -0.044, so the decay takes centuries. Above the threshold (E, F) the intersection lies left of the peak, E^* repels, and the orbit settles onto a sustained cycle that circulates counter-clockwise, with lynx peaks lagging hare peaks.

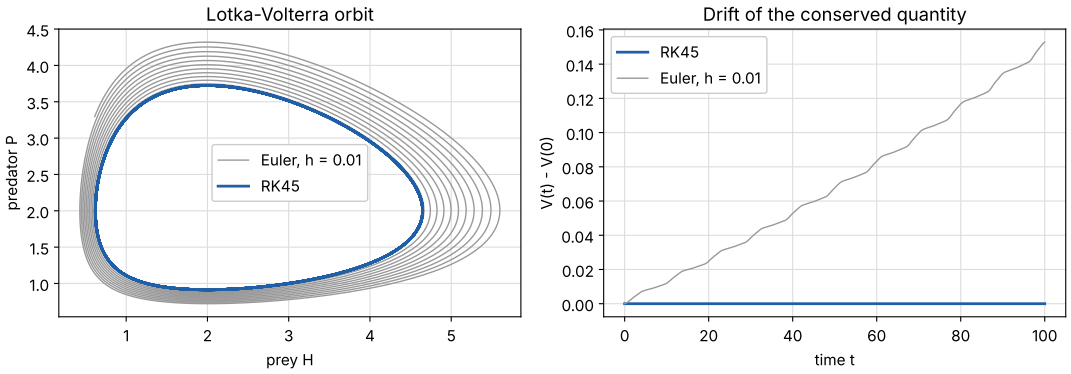


Figure 4: Solver check on Lotka–Volterra. RK45 preserves the conserved quantity V and closes the orbit; Euler ($h = 0.01$) drifts and spirals outward. Only an accurate solver can distinguish a genuine limit cycle from a numerical artefact.

Table 4: Simulated long-run behaviour for the regimes of Section 7. Ranges are over 400 years after the burn-in; period and lynx lag are in years.

	K	Eigenvalues	H range	P range	Period	Lag
A	25	$-1.5,\ -0.073$ (E_1)	25.0	0	–	–
B	60	$-0.188 \pm 0.515i$	30.0	11.25	–	–
C	85	$-0.022 \pm 0.623i$	30.0	14.56	–	–
D	90	$\pm 0.632i$	$[28.5, 31.6]$	$[14.51, 15.49]$	9.94	2.50
E	95	$+0.020 \pm 0.640i$	$[15.8, 50.9]$	$[9.81, 20.75]$	10.25	2.81
F	150	$+0.150 \pm 0.676i$	$[1.1, 134.3]$	$[1.69, 41.16]$	13.91	4.52

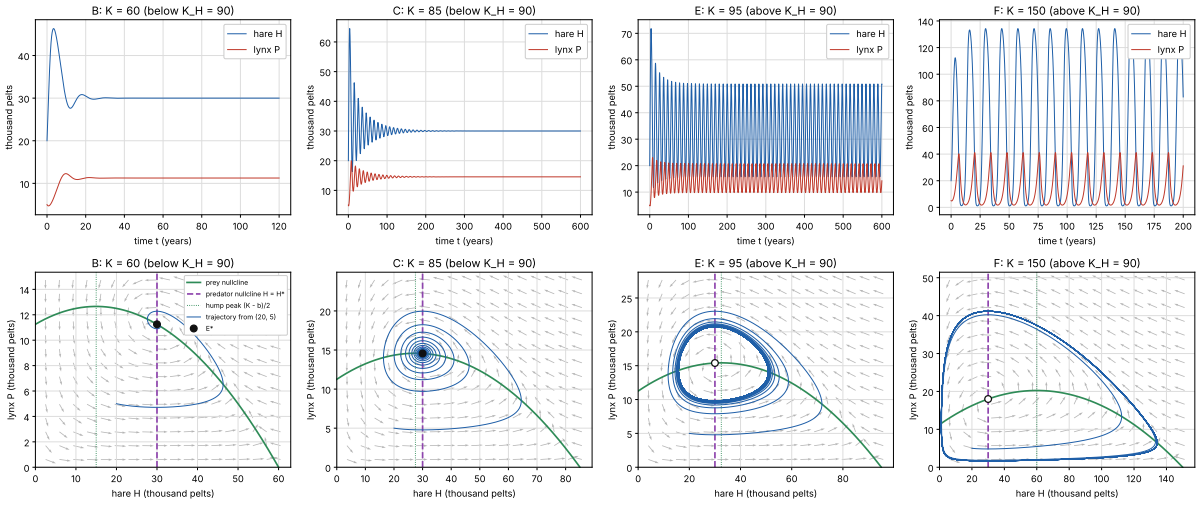


Figure 5: Time series (top) and phase portraits (bottom) from $(20, 5)$ for $K = 60, 85$ (below $K_H = 90$) and $K = 95, 150$ (above). Green: prey nullcline; dashed purple: predator nullcline $H = H^*$; dotted: hump peak $(K - b)/2$. Filled/open circle: stable/unstable E^*. Oscillations decay below K_H and persist above it.

8.4 The limit cycle attracts

For Lotka–Volterra every initial condition lies on its own closed orbit. At $K = 150$ we start one trajectory beside the unstable equilibrium and one far outside the cycle (Figure 6). The first spirals outward, the second inward, and their successive hare maxima converge to the same value, 134.277. The cycle is therefore orbitally stable, in agreement with the uniqueness result cited in Section 7 [4].

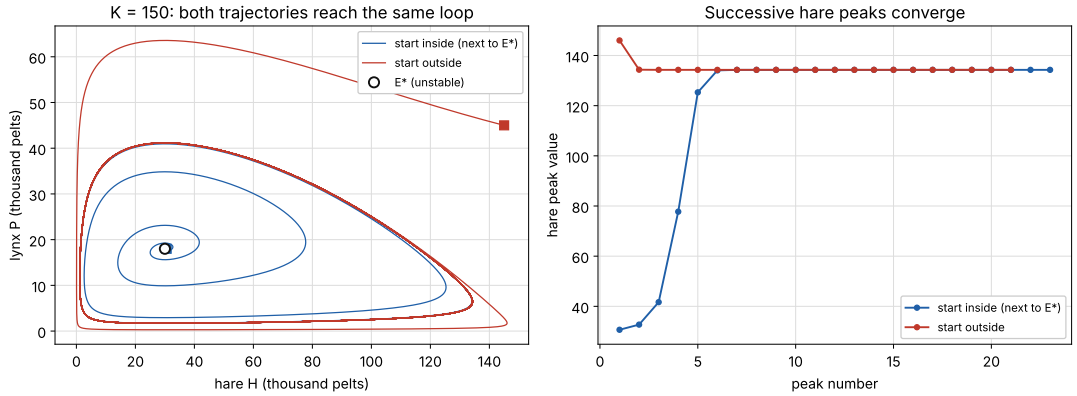


Figure 6: At $K = 150$, trajectories started inside and outside the cycle converge onto the same closed orbit (left); their successive hare maxima approach a common value (right).

8.5 Bifurcation in K

Root-finding on the numerically evaluated trace of $J(E^*)$ places the sign change at $K = 90.0000000000$, in agreement with $K_H = b + 2H^*$ and Figure 2; there the eigenvalues are $\pm 0.632i = \pm i\omega_0$, and they become complex (node to focus) at $K \approx 42.6$.

The numerical bifurcation diagram (Figure 7) records the post-transient maximum and minimum for 113 values of K between 32 and 200. For $K < 89$ the two coincide (largest spread 5×10^{-5}): the solution settles on E^*. At K_H the branches separate continuously from the equilibrium and the cycle grows with K. Just above the threshold its half-amplitude follows $A \approx 7.8\sqrt{K - K_H}$, with fitted log–log slope 0.507 for $0.5 \leq K - K_H \leq 8$ (Figure 8), and the period approaches $T_0 = 9.93$ years (9.97 years at $K - K_H = 0.5$). This square-root growth from zero amplitude is the signature of a supercritical Hopf bifurcation, consistent with the negative first Lyapunov coefficient of Section 7. As K increases further the cycle minima approach zero: at $K = 200$ the hare minimum is 0.12 thousand pelts, the paradox of enrichment [16].

8.6 Where the historical fits lie

We now evaluate the same quantities at every parameter set fitted to the 1900–1915 pelt series in the historical sections: 13 PINN runs (three seeds for each error model, the wider-bounds runs and the physics-weight run) and all 18 matched least-squares starts. Only saved parameters and the training file are read; the held-out years are not used. As a check that the parameters are interpreted exactly as in fitting, re-integrating the selected PINN fit with `model.solve` and the historical solver settings reproduces the saved 1900–1920 trajectory to within 5×10^{-12}. For each set we compute $\sigma = K/K_H$, the eigenvalues $\lambda = \mu \pm i\omega$ of $J(E^*)$, the e-folding time $1/|\mu|$ and the linear period $2\pi/\omega$. We also record the period and lynx lag of the first century simulated from the fitted 1900 state, and flag coefficients within 0.5% of a box limit. Five least-squares starts end in local minima with a training error at least 1.5 times the best start of their variant; they are shown but not interpreted. The complete table is `results/simulations/historical/fits.csv`.

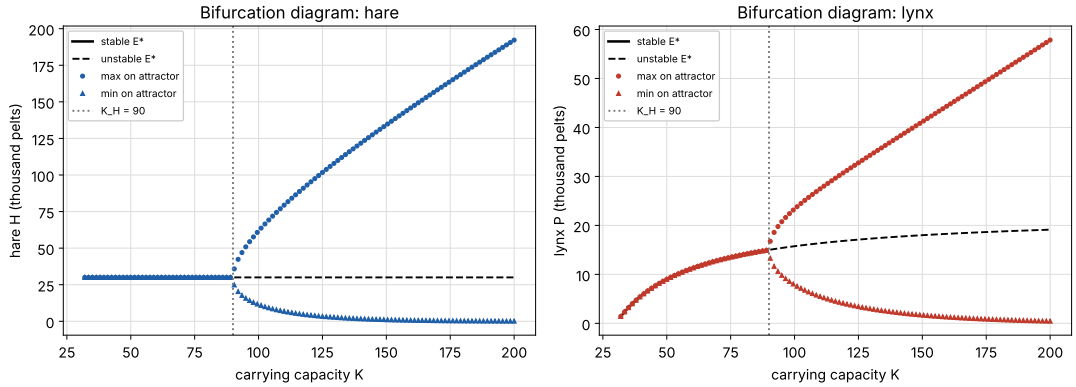


Figure 7: Numerical bifurcation diagram: maximum and minimum of H (left) and P (right) after the burn-in, against K. Solid/dashed black: stable/unstable E^*. A cycle is born at $K_H = 90$ with zero amplitude and its minima approach zero as K grows.

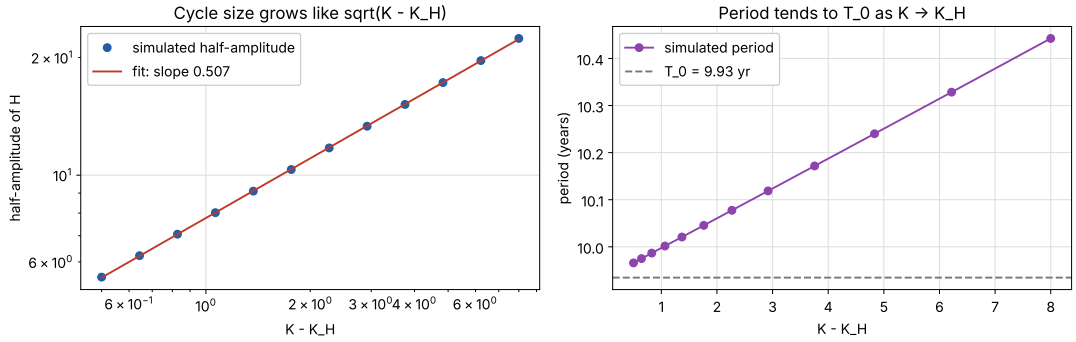


Figure 8: Left: half-amplitude of H against $K - K_H$ on log–log axes, with slope 0.507 (supercritical prediction 1/2). Right: cycle period, tending to $T_0 = 2\pi/\omega_0$.

Table 5: Selected historical fits (lowest training objective within each variant). $\mu \pm i\omega$: eigenvalues at E^*; periods and lag in years; "bounds": coefficients within 0.5% of their box limit.

Method	Variant	σ	$\mu \pm i\omega$	$1/\lvert\mu\rvert$	$2\pi/\omega$	Period, lag	Bounds
PINN	main	1.363	$+0.0079 \pm 0.712i$	127	8.8	10.7, 1.4	K, b upper
PINN	log errors	1.343	$+0.0079 \pm 0.767i$	127	8.2	9.9, 1.3	–
PINN	fitted H_0, P_0	1.347	$+0.0070 \pm 0.699i$	143	9.0	10.7, 1.4	b upper
PINN	wider bounds	1.480	$+0.0049 \pm 0.709i$	204	8.9	10.5, 1.3	b upper
PINN	physics $\times 10$	1.364	$+0.0091 \pm 0.687i$	111	9.1	11.3, 1.4	–
Least squares	main	1.067	$+0.0018 \pm 0.683i$	554	9.2	10.4, 1.6	a, b upper
Least squares	log errors	0.550	$-0.0190 \pm 0.680i$	53	9.2	9.4, 2.0	b upper
Least squares	fitted H_0, P_0	0.993	$-0.0002 \pm 0.669i$	5600	9.4	10.3, 1.7	b upper

Table 5 and Figure 9 show three findings. First, every well-fitting set reproduces the qualitative features of the data: an oscillation with a period of about 10 years and a lynx lag of 1.3–2 years. Second, these sets disagree about the side of the threshold. All PINN fits give σ between 1.32 and 1.48, whereas the least-squares fits give 1.07, 0.99 and 0.55 for the main, fitted-initial-state and log-error variants. Third, the half-saturation constant b lies on its upper box limit in nearly every fit, and widening the box moves the PINN estimate from $\sigma \approx 1.36$ to 1.48, so the reported σ is conditional on the bounds.

The reason the data cannot settle the question is the size of μ. In every well-fitting set $|\mu|$ is between 0.0002 and 0.019 per year, so oscillations grow or decay by a factor e only over 53 to 5600 years. Over the 16 training years the factor $e^{16\mu}$ lies between about 0.74 and 1.16, and a slowly damped focus and a slowly growing spiral produce almost the same two cycles. Figure 10 shows the bifurcation diagram in K for the two selected main fits, with all other coefficients held at their fitted values. For the least-squares fit the long-run cycle (H between 5.6 and 85 thousand pelts, period 10.8 years) resembles the observed range. For the PINN fit the long-run cycle is much larger (H between 0.1 and 247, period 18.5 years); the fitted trajectory over 1900–1920 is still an early, slowly growing part of that spiral.

We therefore do not conclude that the historical system lies on a stable limit cycle. The fitted model is close to the Hopf threshold, oscillates with the observed period and lag, and which side of the threshold it falls on depends on the fitting method, the error model and the bounds.

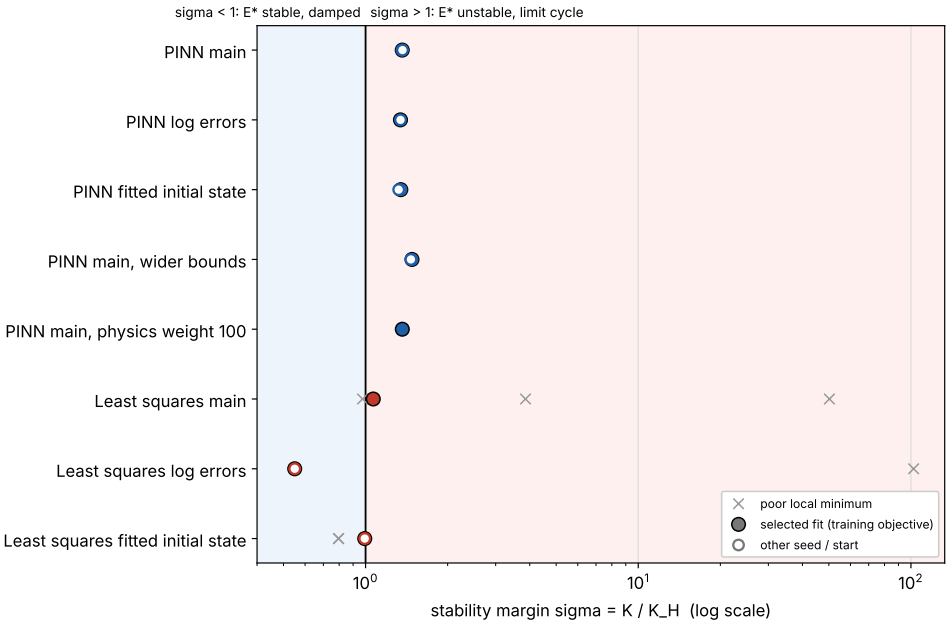


Figure 9: Stability margin $\sigma = K/K_H$ of every fitted parameter set (log scale). Filled: selected fit of each variant; open: other seeds or starts; crosses: least-squares local minima with at least 1.5 times the best training error. PINN fits lie at $\sigma \approx 1.3$–1.5; least-squares fits straddle $\sigma = 1$.

8.7 Scope

The illustrative simulations provide numerical evidence for one parameter set, not proofs. Near K_H the results depend on the burn-in length, which is why it is lengthened there. The historical placement uses the saved fits as given; their σ values inherit the bounds, the error model and the observation convention of the fitting sections, and the pelt series are abundance indices rather than population counts.

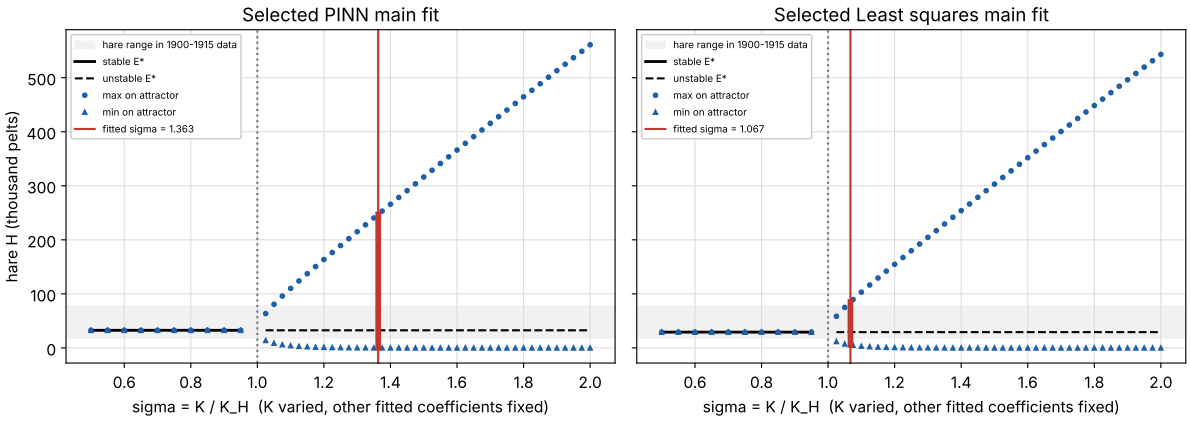


Figure 10: Bifurcation diagram in K for the selected PINN (left) and least-squares (right) main fits, plotted against $\sigma = K/K_H$ with the other fitted coefficients fixed. Red line: fitted σ; thick red bar: long-run hare range at the fitted parameters; grey band: hare range in the 1900–1915 training data. Near $\sigma = 1$ transients decay very slowly, so points there still carry residual oscillation after the burn-in (at most 60,000 years).

9 Historical inverse PINN study

This is Yash's contribution, using the historical pelt series supplied by Advay. It extends the separately preserved synthetic recovery experiment; its individual scope is retained alongside the combined group deliverables. The Stan compilation contains annual Hudson's Bay Company hare and lynx pelt returns [3]. We train on 1900–1915 (16 pairs) and evaluate once on frozen forecasts for 1916–1920 (five pairs). Counts are thousands of pelts, treated as abundance indices rather than censuses. Annual totals are approximated as point samples. No observation scale factor is fitted. Alternative source compilations disagree; all results here are conditional on the Stan series.

9.1 Model and protocol

With $t = \text{Year} - 1900$, the shared dimensional model is

$$\dot{H} = rH(1 - H/K) - aHP/(b + H), \qquad \dot{P} = eaHP/(b + H) - mP.$$

All six positive coefficients are inferred. Bounds below are exploratory numerical choices, not measured biological ranges. No coexistence constraint is imposed and e is not constrained below one because these are effective coefficients in pelt units. The starting vector is $(r, K, a, b, e, m) = (0.8, 150, 2, 30, 0.6, 0.6)$. The shared historical configuration records this protocol and its exploratory bounds. The local least-squares benchmark uses the identical config, data split, scaling and solver.

Coefficient	Bounds	Selected primary estimate
r	0.05–3	0.60546
K	20–500	497.6
a	0.05–10	9.2605
b	0.5–300	299.98
e	0.02–5	1.0966
m	0.05–3	0.99388

9.2 Network, loss and optimization

The inverse PINN [14] uses three hidden layers of 48 tanh units, float64, softplus-positive normalized outputs, and sigmoid-bounded coefficients. Species scales are training maxima $(s_H, s_P) = (77.4, 59.4)$; network time is $2t/15 - 1$. For $u_\theta = (H_\theta/s_H, P_\theta/s_P)$, automatic differentiation evaluates $R_\theta = \partial_t u_\theta - f(u_\theta \odot s; p)/s$. The loss is

$$L = \text{mean}(u_\theta - y/s)^2 + 10\,\text{mean}\,R_\theta^2 + 10\,\text{mean}(u_\theta(0) - y_0/s)^2.$$

The initial observation is $y_0 = (30, 4)$. There are 161 collocation times in $[0, 15]$. Optimization uses 500 data/IC warmup steps, up to 5000 joint Adam steps at learning rate 0.002 (gradient norm clipped at 10), then up to 600 L-BFGS iterations with strong-Wolfe line search. Training loss is logged every 100 Adam steps; plateau stopping uses 15 checks and minimum improvement 10^{-8}. The best recorded weighted training objective determines the saved state. Seeds are 17, 29 and 43, with one CPU thread and deterministic Torch operations. Each variant selects its own lowest-loss seed; objectives from different observation models are not used to select the primary analysis.

9.3 Frozen forecasts and matched benchmark

The selected primary seed is 17. Forecasts use the learned coefficients in the shared RK45 solver with rtol 10^{-8}, atol 10^{-10} and maximum step 0.1 year. Integration starts from $(30, 4)$ in 1900

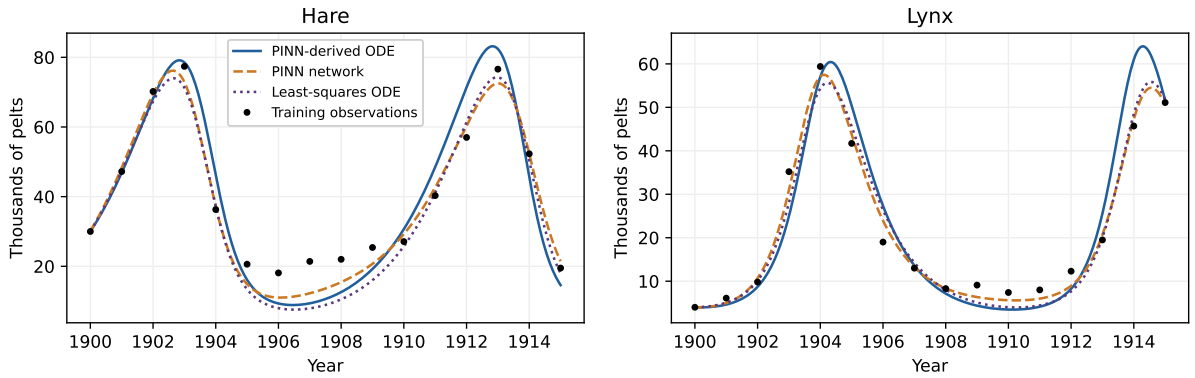


Figure 11: Training observations, learned network and re-integrated ODE. A small local physics residual does not guarantee identical long-horizon ODE and network trajectories.

and proceeds through 1920, with no restart from a test state. The neural network is never extrapolated for forecasting. Its starting-state discrepancy is $(0.10060, -0.00723)$ thousand pelts; the ODE still starts at the exact observation. The primary ODE training normalized MSE is 0.00981. The matched primary least-squares fit uses six starts (RNG seed 91), bounded `least_squares`, up to 500 function evaluations and ftol=xtol=gtol=10^{-10}. Each start and its convergence status is retained. This independent benchmark was subsequently reproduced by Advay on the same configuration; the combined report documents the chronology. PINN and least squares differ in soft versus exact IC enforcement and in the PINN's added physics penalty.

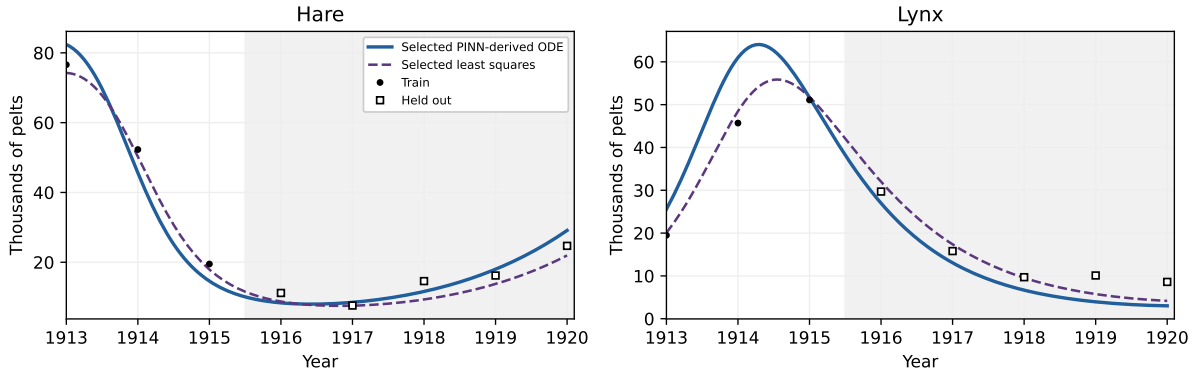


Figure 12: Frozen forecasts. Thin pale lines are the three primary PINN seeds, not confidence bands. Shading marks held-out years; observations were used for scoring only after all forecast and seed choices were frozen.

	RMSE		MAE	
Method	Hare	Lynx	Hare	Lynx
PINN fitted IC	3.687	4.697	3.057	4.530
PINN log errors	18.351	10.026	14.289	8.972
PINN primary	2.847	4.317	2.590	4.046
PINN wider bounds	2.976	4.053	2.721	3.779
PINN physics 100	4.519	3.978	4.107	3.496
LS primary	3.072	3.043	2.586	2.559
LS log errors	6.874	1.596	6.626	1.381
LS fitted IC	2.246	3.236	2.083	2.713

All scores are thousands of pelts on the same five held-out years. The primary PINN has slightly

lower hare RMSE than primary least squares, but higher lynx RMSE. Hare MAE is essentially tied. These results do not establish an overall PINN advantage.

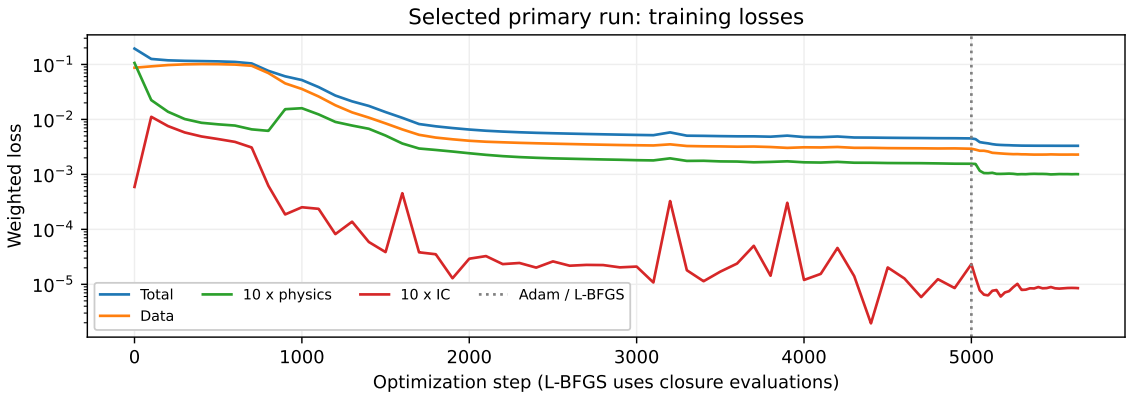


Figure 13: Selected primary training loss components. L-BFGS abscissae count closure evaluations after Adam; they are not directly comparable computational steps.

9.4 Sensitivity and practical identifiability

All sensitivities were specified before any test scoring and are reported regardless of outcome. The main three seeds were repeated with log observation residuals, and separately with trainable initial states in $[0.5, 1.5]$ times the 1900 observations. The first data point stays in the data loss. The fitted-IC PINN's ODE starts at approximately $(34.7477, 3.7841)$; the network and ODE initial states are separately recorded. The benchmark repeats both observation/IC sensitivities with six starts each. Log residuals worsen hare forecast error for both methods, especially the PINN. None of these scores changes the predeclared primary choice.

To check arbitrary box limits, each lower bound was halved and each upper bound doubled, and the main PINN was repeated for all three seeds. A further seed-17 run increased physics weight from 10 to 100. In total there are 13 PINN runs and 18 least-squares starts. The wider-box fits move K from about 498 to about 986 and b from about 300 to about 600, close to the expanded limits; a also shifts substantially. Similar results across seeds therefore do not demonstrate uniquely recovered biological parameters. The wider-box forecast changes are smaller than these coefficient shifts.

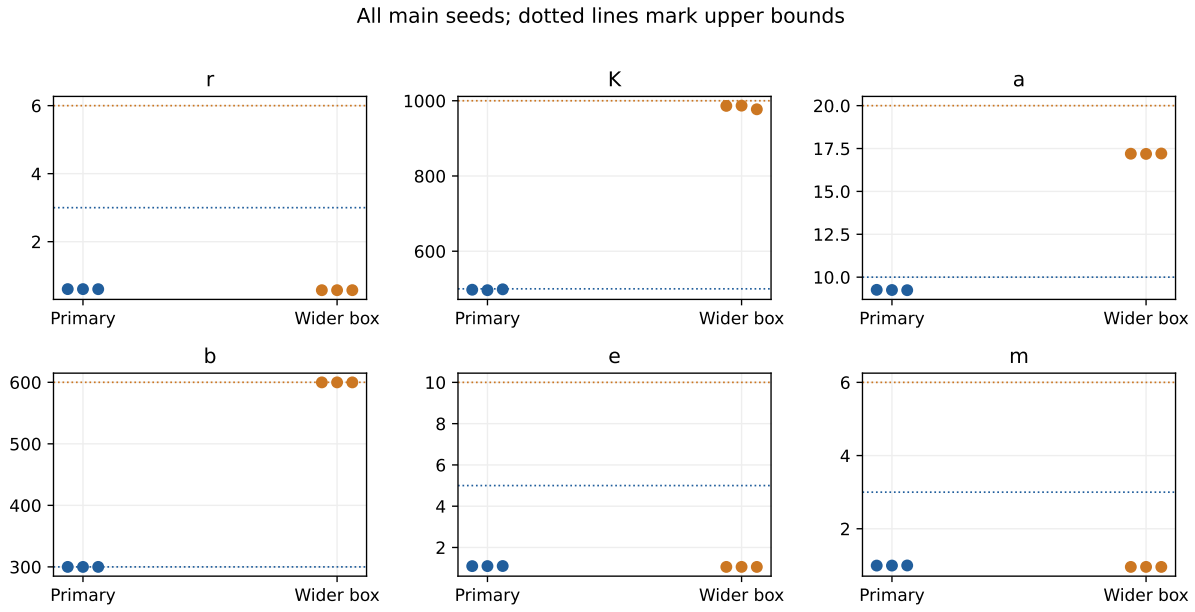


Figure 14: Primary versus wider-box parameter estimates over three seeds. Several effective coefficients track the numerical bounds; narrow seed spread is not parameter certainty.

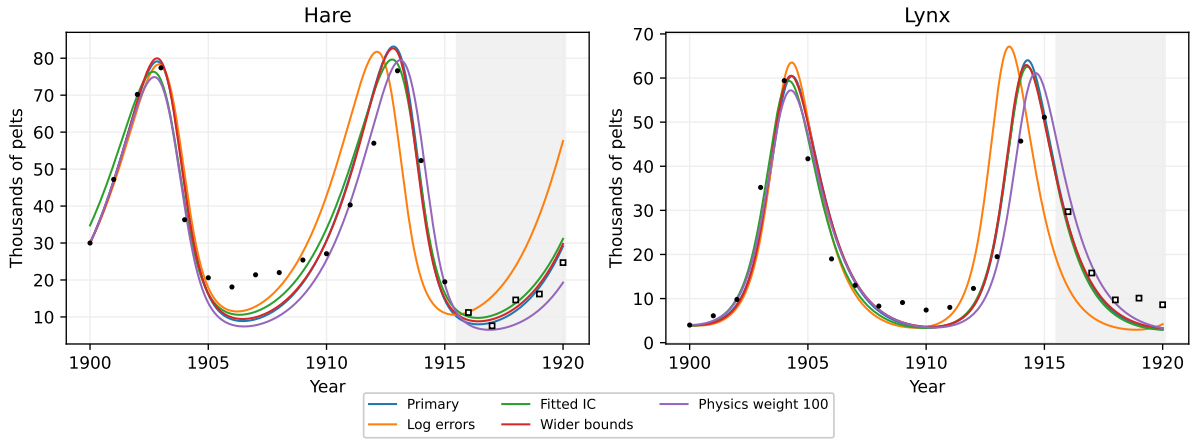


Figure 15: All selected PINN variants. The log-error fit shows appreciable model/ODE forecast sensitivity. Variants were retained rather than chosen using held-out errors.

The local trajectory sensitivity matrix uses central differences with respect to log parameters (step 10^{-4}), normalized using the declared observation loss. Its selected primary condition number is approximately 317.5. This is a local diagnostic, not a proof of structural identifiability or a confidence interval. For feasible coexistence, $H^* = mb/(ea - m)$, $K_H = b + 2H^*$ and $\sigma = K/K_H$; the selected primary fit gives $\sigma = 1.363$. This fitted ODE lies on the cycling side of Joel's threshold, conditional on the model and protocol. It does not prove that the historical ecological system has a stable limit cycle. All seeds' values and feasibility flags are exported to the team.

9.5 Reproducibility and limitations

`configs/historical.json` records every primary choice; the two sensitivity configs record the expanded box and physics weight. Run manifests hash sources, config, training data, forecasts and diagnostics. A forecast-freeze record is written before the test observations are read. `scripts/run_historical_pinn.py` accepts only the training table; `scripts/score_historical.py` handles final scoring. Existing synthetic recovery results are preserved as a separate controlled validation.

Six effective coefficients are fitted from roughly two noisy cycles; boundary hits, source disagreement, annual season totals, unknown catchability and the small test window limit interpretation. PINN physics is a soft constraint, and the log-loss variant exposes an appreciable network/ODE discrepancy. Seed spread is optimization sensitivity only. Forecast skill on five points cannot establish general superiority, causal mechanisms or absolute animal populations. This report and notebook concern Yash's component; the accompanying combined notebook and report integrate all five contributions.

PINN optimization gap. Advay independently evaluated the training observation objective on the frozen PINN-derived ODE trajectories. Their costs are 1.859, 3.396 and 1.866 times the least-squares minima for main, log-error and fitted-initial-state variants. These are observation costs, not the PINN's weighted neural training loss. The PINN optimizes a soft-physics network objective, and reintegration can accumulate phase error; it does not attain the least-squares optimum here. These diagnostics motivate the reported limitations and do not trigger post-test retuning.

10 Data, Least-Squares Baseline and Validation

This section states what the historical observations are, how they were prepared, how the model was fitted to them by nonlinear least squares, and how well the fitted model forecasts years it never saw. Every analysis choice that could be influenced by the held-out years was written down and committed to the project repository before those years were opened.

10.1 Data source and meaning

We use annual counts of snowshoe hare and Canadian lynx pelts traded by the Hudson's Bay Company, 1900–1920, in thousands of pelts, as distributed with the Stan case study [3]. Carpenter traces the numbers through Howard's lecture notes [8] to a biology textbook; the original source is the Company's fur returns, discussed by Hewitt [6]. The values are used unchanged (`data/historical/pelts_1900_1920.csv`); the file's provenance, checksums and every transformation (removing comment lines, reordering columns) are recorded alongside it.

These are *trade counts*, not censuses. Trapping effort, fur prices, the number of reporting posts and animal behaviour all affect them; Hewitt notes that hungry lynx are easier to trap when hares are scarce. A second widely used compilation, the tsibbledata `pelt` series [13], differs noticeably over the same years (Figure 16): the correlation between the two is 0.81 for lynx but only 0.32 for hare, and individual hare values differ by more than our forecast errors. We use the Stan series because the project protocol names it, and treat the disagreement as a source of uncertainty that no model in this report can remove.

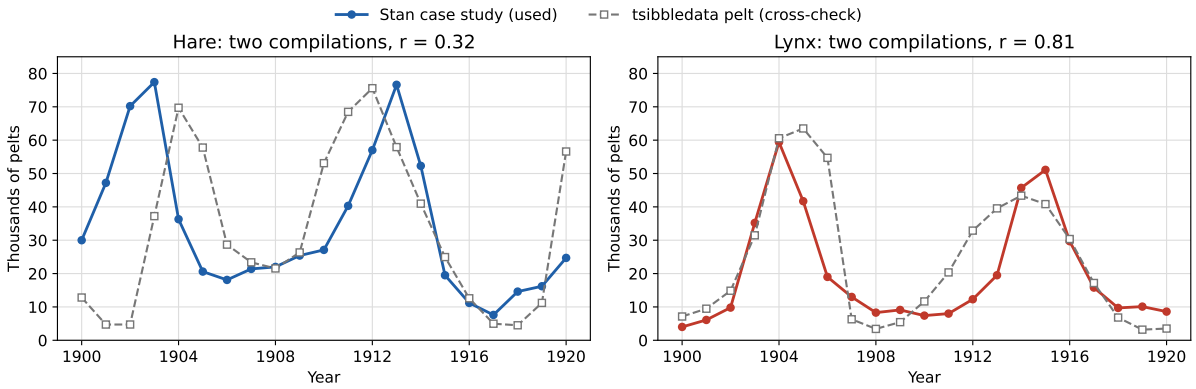


Figure 16: The Stan case-study series used here (solid) and the tsibbledata `pelt` compilation (dashed) for 1900–1920. Both show two cycles, but yearly values differ, especially for hare.

Observation model. Following the agreed convention (assumption O1), pelt counts are taken to be proportional to animal numbers with unknown constant factors, and the model states are defined directly in pelt units:

$$Y_{H,i} = H(t_i) + \varepsilon_{H,i}, \qquad Y_{P,i} = P(t_i) + \varepsilon_{P,i}, \qquad t_i = \text{Year}_i - 1900,$$

with each annual total treated as a sample of the state at t_i. No observation scale factor is fitted. Under O1, r, m and the dimensionless groups of the model section are unchanged by the unknown factors; K, b, a and e are *effective* coefficients in pelt units and are not interpreted as absolute biological quantities.

10.2 Preparation and exploratory checks

The 21 rows are consecutive, with no missing values, no zeros and no duplicates. The data were split once into a training file (1900–1915, 16 rows) and a test file (1916–1920, 5 rows), with

checksums recorded; all fitting code reads only the training file. Every statistic below uses the training years only.

- **Cycles and lag.** Hare peaks in 1903 and 1913 (a ten-year spacing); lynx peaks in 1904 and at the 1915 boundary. The correlation of $H(t)$ with $P(t+k)$ is largest at $k = 2$ years ($r = 0.94$; 0.76 at $k = 1$). In the hare–lynx plane the data trace counter-clockwise loops, as predator–prey models predict (Figure 17).
- **Outliers.** None: robust z-scores of year-to-year log growth stay below 3 (largest 2.94, the 1903–1904 hare crash).
- **Scale.** Training ranges are 18.1–77.4 (hare) and 4.0–59.4 (lynx). Hare counts are about twice lynx counts, so residuals are divided by each species' training maximum $(s_H, s_P) = (77.4, 59.4)$ to weight the species equally. No log transform is used in the main analysis; a log-residual fit is a declared sensitivity check.
- **Split position.** Training ends exactly at the 1915 lynx peak, so the test years are the following crash and the start of recovery: a turning point the fit never sees in that cycle.

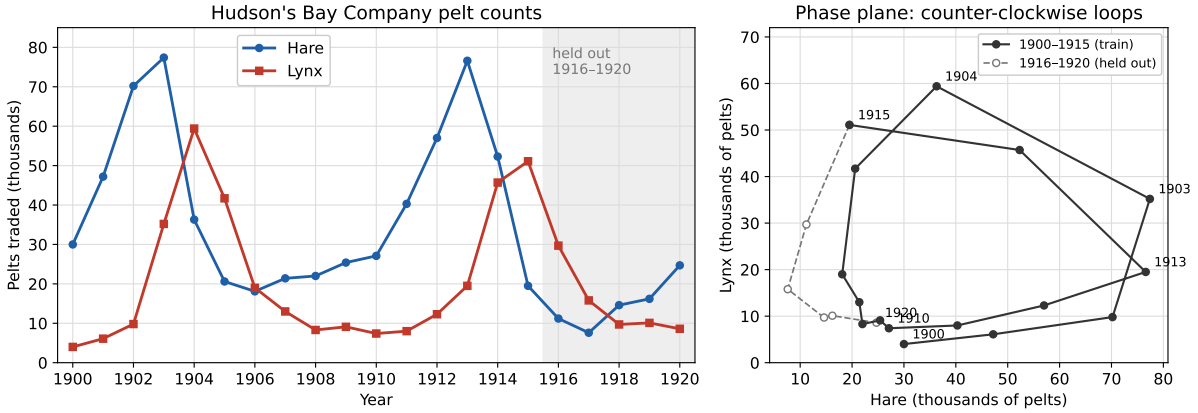


Figure 17: Left: annual pelt counts, 1900–1920; the shaded band is held out. Right: the same data in the hare–lynx plane (training solid, held-out dashed).

10.3 Least-squares baseline

Problem. With $\theta = (r, K, a, b, e, m)$ and the shared RK45 solver (relative tolerance 10^{-8}, absolute 10^{-10}, maximum step 0.1 years), let $\hat{H}(t;\theta)$, $\hat{P}(t;\theta)$ solve the model from the observed 1900 state $(30.0, 4.0)$. The main estimate minimises

$$C(\theta) = \tfrac{1}{2} \sum_{i=1900}^{1915} \left[\left(\frac{\hat{H}(t_i;\theta) - Y_{H,i}}{s_H} \right)^2 + \left(\frac{\hat{P}(t_i;\theta) - Y_{P,i}}{s_P} \right)^2 \right]$$

over the shared box bounds $r, m \in [0.05, 3]$, $K \in [20, 500]$, $a \in [0.05, 10]$, $b \in [0.5, 300]$, $e \in [0.02, 5]$. Two declared sensitivity variants change one assumption each: *log errors* replaces the scaled residuals by $\log \hat{Y} - \log Y$, and *fitted 1900 state* adds (H_0, P_0) as unknowns within $[0.5, 1.5]$ times the observed values.

Method. The bounded trust-region reflective algorithm (SciPy `least_squares`) is run in log-parameter space $x = \log \theta$, because the bounds span up to a 600-fold range. Finite-difference steps of 10^{-4} in x stay well above the solver tolerance. The objective is multimodal, so each variant uses 80 starts: the shared initial guess, 39 Latin-hypercube points uniform in $\log \theta$ and 39 uniform in θ. The fit with the lowest training cost is selected.

Protocol record. The settings were committed before the first run. That run (40 log-space starts) missed the best *fitted 1900 state* fit, which an independent benchmark had found at training cost 0.063 against our 0.70; evaluating our objective at the benchmark's parameters gave 0.063, so the code agreed and the start design was at fault (the optimum sits near the upper bounds, which log-uniform starts rarely visit). The uniform-scale starts were added as a declared amendment, judged on training cost alone and before any test score of the baseline was computed. Both runs are kept.

Variant	C	r	K	a	b	e	m	$\sigma = K/K_H$
Main	0.0845	0.647	382	10.0*	300*	0.967	0.857	1.07 (cycle)
Log errors	0.903	0.520	197	7.17	300*	1.80	1.15	0.55 (damped)
Fitted 1900 state	0.0634	0.520	362	8.15	300*	1.32	1.05	0.99 (damped)

Table 6: Least-squares estimates. * at the upper bound. Units: r, m per year; K, b thousands of hare pelts; a, e effective pelt-unit coefficients. The fitted 1900 state in the last variant is $(37.1, 3.64)$.

Results. Table 6 and Figure 18 give the selected fits. Three findings matter for interpretation.

1. *The optimum is computationally well defined.* In every variant all starts within 1% of the best cost (18, 42 and 16 of 80) found the same parameters to four significant figures, and a separately written least-squares code by another team member reaches the same optimum (training costs within 2×10^{-7}, forecasts within 0.002 thousand pelts).

2. *The saturation of predation is not identified.* b sits on its upper bound in every variant. With b about four times the largest hare count, $aH/(b + H) \approx (a/b)H$ over the data: predation is nearly linear in hare abundance. In the linearised fit the correlation between $\log a$ and $\log b$ is 0.998 and the approximate standard errors of $\log K$, $\log a$, $\log b$ exceed 1 (a factor of three or more), whereas r, e, m are determined to about 10–40%. Only the ratio $a/b \approx 0.024$–0.033 is well determined. Because the optimum lies on the boundary, its location depends on the chosen box.

3. *Limit cycle or damped oscillation is not decided.* Using the Hopf threshold $K_H = b(ea + m)/(ea - m)$ from the stability analysis, σ is 1.07 for the main fit (just past the threshold), 0.99 with a fitted 1900 state (just below) and 0.55 with log errors. All three reproduce the observed timing equally well (hare peaks fitted at 1902.5–1902.8 and 1912.6–1913.0, lynx 1.6 years later, period about ten years). Two cycles of data cannot distinguish a stable limit cycle from a slowly damped oscillation (Figure 19).

10.4 Held-out evaluation, 1916–1920

Protocol. A scoring plan fixing the forecasts compared, the headline comparison, the metrics and the interpretation rule was committed and published before the test file was opened. All forecasts integrate frozen parameters from 1900 through 1920 with the shared solver; nothing restarts from a test value, and no neural-network extrapolation is used. For each species separately we report RMSE $= \left(\frac{1}{5}\sum(\hat{Y} - Y)^2\right)^{1/2}$ and MAE $= \frac{1}{5}\sum|\hat{Y} - Y|$ in thousands of pelts. With five test points a method is called *better* only if both its RMSE and MAE are smaller *and* it is closer in at least three of the five years; otherwise the methods are *similar*. Two reference forecasts use training data only: the 1915 value held constant (persistence) and the 1900–1915 mean. The headline comparison is the main least-squares fit against the PINN-derived ODE forecast of the historical PINN study (seed chosen by training loss before scoring).

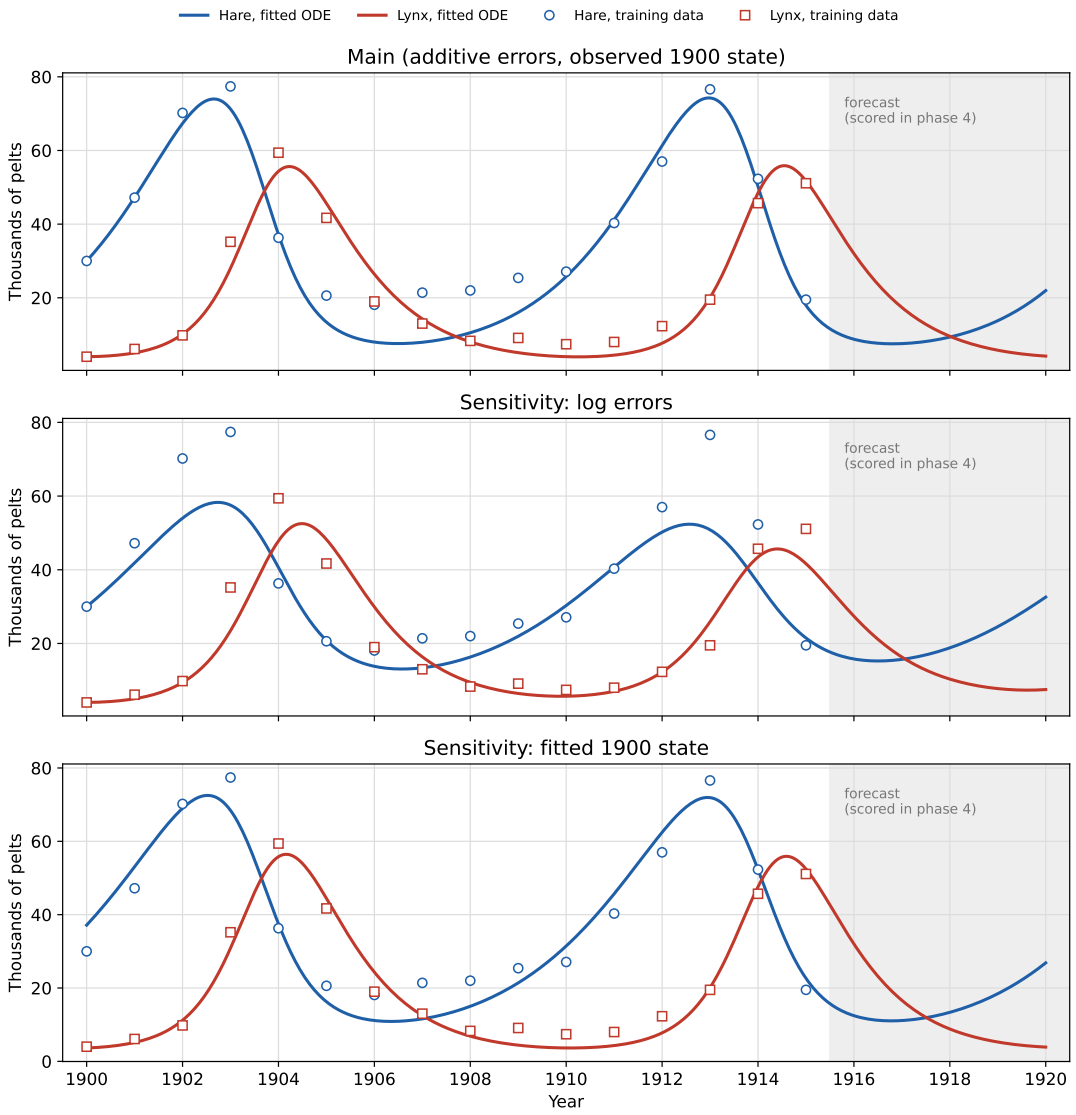


Figure 18: Least-squares fits to the 1900–1915 pelt counts (markers) and their continuation through 1920 (shaded, held out) for the main analysis and both sensitivity variants.

Every frozen forecast was checksum-verified, and the scores reproduce the PINN study's own evaluation exactly.

Variant	Method	Hare RMSE	Hare MAE	Lynx RMSE	Lynx MAE
Main (headline)	Least squares	3.07	2.59	3.04	2.56
	PINN-derived ODE (seed17)	2.85	2.59	4.32	4.05
Log errors	Least squares	6.87	6.63	1.60	1.38
	PINN-derived ODE (seed29)	18.35	14.29	10.03	8.97
Fitted 1900 state	Least squares	2.25	2.08	3.24	2.71
	PINN-derived ODE (seed17)	3.69	3.06	4.70	4.53
Reference	Persistence (1915 value)	7.38	6.72	37.16	36.32
Reference	Training mean	25.87	25.23	10.58	10.21

Table 7: Held-out errors, 1916–1920, thousands of pelts.

Results. (Table 7, Figures 20 and 21.)

- **Both methods forecast the crash and recovery.** In the main variant both predict

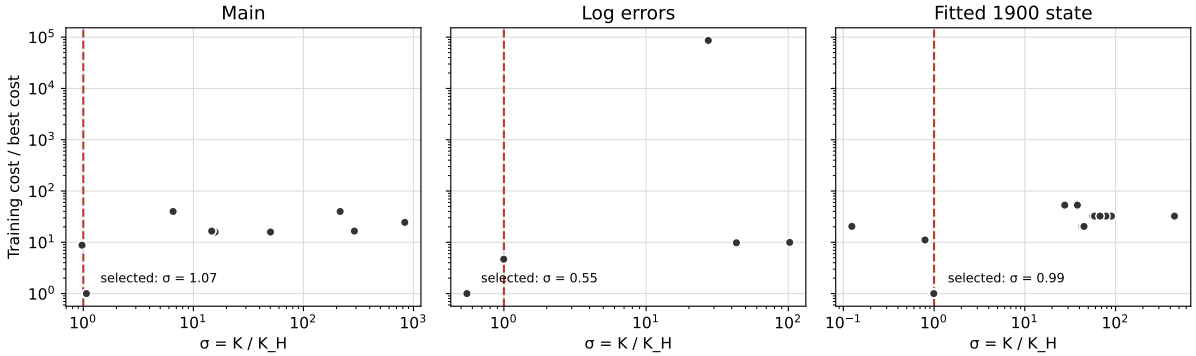


Figure 19: Training cost relative to the best, against $\sigma = K/K_H$, for every converged start. The dashed line is the Hopf threshold $\sigma = 1$. The best fits sit near $\sigma = 1$, on different sides depending on the variant.

the 1916–1920 lynx collapse and the hare recovery to within about 3–4 thousand pelts, far better than persistence (lynx RMSE 37.2) or the training mean (lynx 10.6, hare 25.9). The fitted dynamics carry real information about the next half-cycle.

- **Headline: mixed and close.** Hare: similar (RMSE 3.07 least squares, 2.85 PINN; MAE 2.59 both). Lynx: least squares better (3.04 against 4.32, closer in all five years); the PINN-derived lynx forecast falls to 3–5 thousand after 1918 where 9.7–10.1 were observed.

- **Sensitivity variants.** Least squares is better or similar in both. The log-error PINN fit places the 1913 peaks about a year early and its forecast errors are large (hare RMSE 18.4).

- **Why they differ.** Evaluated on C, the selected PINN parameters fit the training years 1.9–3.4 times worse than the least-squares optimum, as expected for an estimator that minimises a combined data and physics loss, and they cycle more strongly ($\sigma \approx 1.35$). A better training fit did not give a better hare forecast in the main variant. With five points, none of these differences is strong evidence.

10.5 Limitations

- **A very small test.** Five annual points, about half a cycle, beginning at the 1915 lynx peak. The comparison measures one crash and recovery, not long-run accuracy.

- **Trade counts, not populations.** O1 assumes constant catchability; trapping effort, prices and lynx behaviour almost certainly varied. Annual totals are treated as point samples, a mismatch of up to a year in timing.

- **Source uncertainty.** Two compilations of the same records disagree by more than the forecast errors in some years; all results are conditional on the Stan series.

- **Initial state.** The main analysis treats the noisy 1900 counts as exact. Fitting them instead moves σ from 1.07 to 0.99 and lowers the hare error.

- **Identifiability.** K, a and b are individually poorly determined and b (and a in the main fit) lies on the box boundary, so their values depend on the chosen bounds. A good curve fit is not evidence of uniquely recovered biological rates.

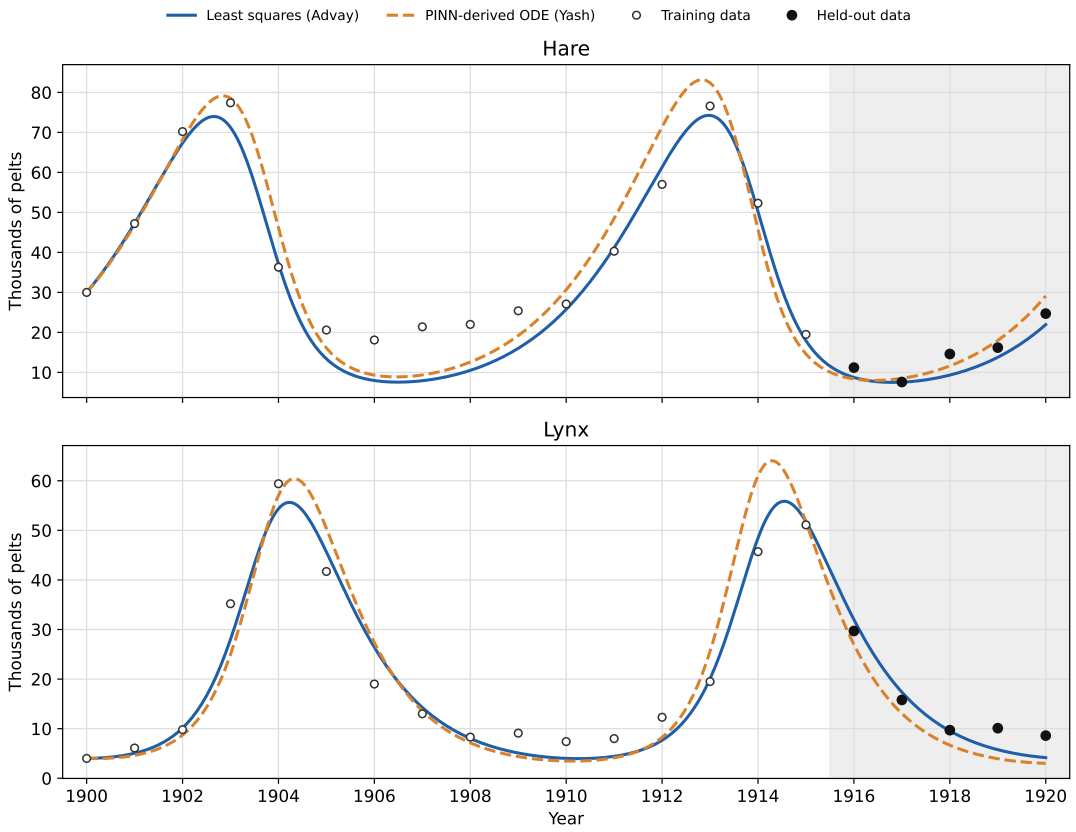


Figure 20: Least-squares (solid) and PINN-derived (dashed) fits, main variant. Open circles: training data; filled circles: held-out data (shaded). Both were fitted to 1900–1915 only and integrated forward from 1900.

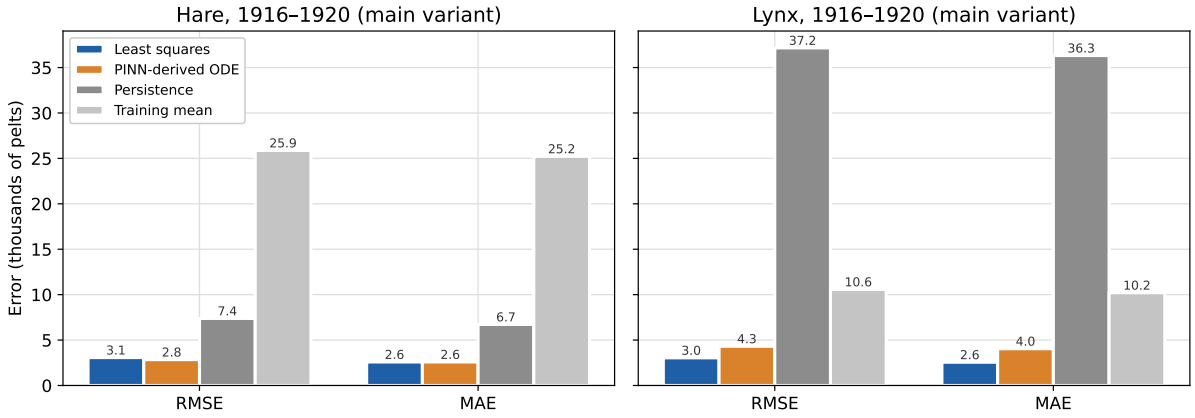


Figure 21: Held-out RMSE and MAE, main variant, with the persistence and training-mean reference forecasts.

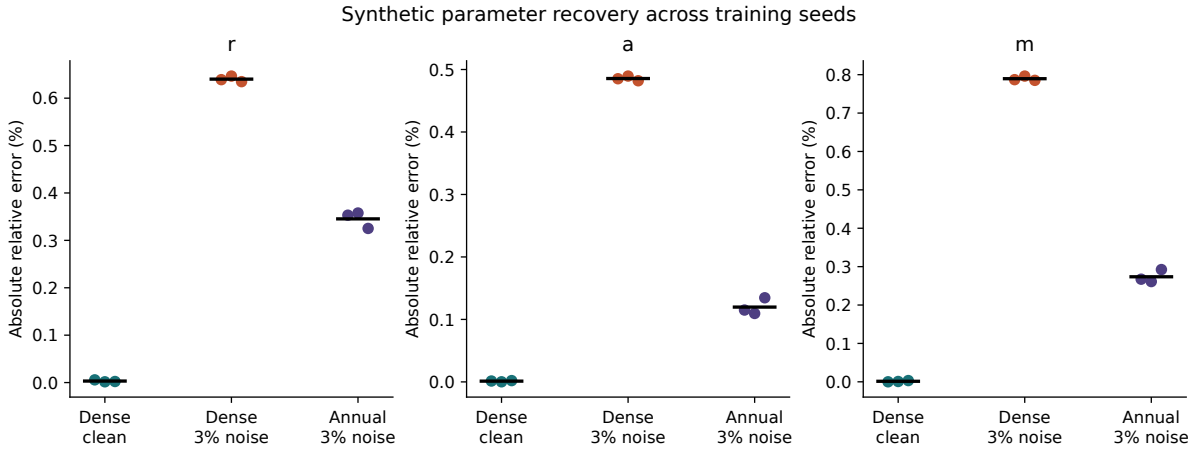
- **Observation model.** Changing only the error model (additive to log) changes σ from 1.07 to 0.55 and the hare/lynx error balance, so the qualitative dynamics inferred are sensitive to how the counts are assumed to be measured.

- **Missing ecology.** A closed two-species, constant-parameter model omits vegetation, other predators, weather, disease and spatial structure. Its success over 1916–1920 shows that it captures the dominant short-term feedback, not that it is complete.

Reproduction. `scripts/advay_data_checks.py`, `scripts/advay_make_split.py`, `scripts/advay_fit_baseline.py` (`-protocol v2`), `scripts/advay_compare_benchmark.py` and `scripts/advay_score_heldout.py` regenerate every number and figure in this section; the declared protocols are in `docs/advay_baseline_protocol.md` and `docs/advay_scoring_plan.md`.

11 Controlled synthetic validation

The separate synthetic study tests implementation recovery with known parameters and 3% observation noise. Across three annual-data seeds, the largest relative error in the three inferred rates is below 0.36%. K, b and e are correctly fixed at generating values for that controlled test; historical fitting instead estimates all six. This favorable recovery experiment is not evidence of historical parameter accuracy. Its complete program, recovery tables, residuals and sensitivity runs remain in `notebooks/yash_pinn.ipynb` and `results/synthetic/`.



Reproduction and acknowledgement. Run the combined notebook `notebooks/group_project.ipynb` from a fresh Python environment using the supplied requirements. Full expensive experiment drivers and individual notebooks remain alongside the frozen evidence. All report plots have EPS sources and vector PDF companions. Per-person AI assistance and actual manual-review records are in `docs/ai_acknowledgement.md` and Arnav's contribution folder. Each member remains responsible for understanding and explaining their own work.

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